Commitment and Randomization in Communication

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Abstract

When does Sender, in a Sender-Receiver game, strictly value commitment? In a setting with finitely many actions and states, we establish that, generically, commitment has no value if and only if a partitional experiment is optimal. Moreover, if Sender's preferred cheap-talk equilibrium necessarily involves randomization, then Sender values commitment. Our results imply that if a school values commitment to a grading policy, then the school necessarily prefers to grade unfairly. We also ask: for what share of preferences does commitment have no value? For any state space, if there are $|A|$ actions, the share is at least $\frac{1}{|A|^{|A|}}$. As the number of states grows large, the share converges precisely to $\frac{1}{|A|^{|A|}}$.

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1 Introduction

Commitment is often valuable. In the context of communication, this fact is brought out by the contrast of Sender's payoff in Bayesian persuasion versus cheap talk. For any prior, and any profile of Sender's and Receiver's preferences, Sender's payoff is always weakly higher under Bayesian persuasion than in any cheap-talk equilibrium.[1] In this paper, we ask: when does commitment make Sender *strictly* better off?

Answering this question would contribute to our understanding of circumstances that incentivize building strong institutions that are immune to influence (North 1993; Lipnowski, Ravid, and Shishkin 2022) or building a reputation for a degree of honesty (Best and Quigley 2024; Mathevet, Pearce, and Stacchetti 2024).

We focus exclusively on environments with finitely many states and actions. We show that, generically, Sender with commitment values that commitment if and only if he values randomization (Theorem 1).[2] In other words, the Bayesian-persuasion payoff is achievable in a cheap-talk equilibrium if and only if a partitional experiment is a solution to the Bayesian-persuasion problem. Moreover, if Sender's preferred equilibrium in a cheap-talk game necessarily involves randomization, then Sender values commitment (Theorem 2).

For an application of these results, consider a school that assigns grades to students, each of whom is characterized by a vector of attributes. Some of the attributes are relevant, in the sense that an employer values those attributes or the school's value of placing a student depends on them. Other attributes are irrelevant. The school assigns a grade to each student based on her attributes. The school's grading policy is *fair* if it assigns the same grade to students with identical relevant attributes. Theorem 1 tells us that if the school values committing to a grading policy of any form (such as mandating a maximum GPA), then the school prefers to grade unfairly. Conversely, if a fair grading scheme is optimal, there is no need for commitment: discretionary "cheap-talk" grades are as effective as those disciplined by a publicly declared grading policy.

We also derive results about the share of preferences such that Sender finds commitment (or,

[1] In fact, Bayesian persuasion provides the upper bound on Sender's equilibrium payoff under any communication protocol, such as disclosure or signaling; see Proposition 1 in the Online Appendix of Kamenica and Gentzkow (2011).

[2] Most of the results in this paper hold generically, i.e., for all but a knife-edge set of preferences. To ease exposition, we henceforth sweep the genericity modifier under the rug in the Introduction.

equivalently, randomization) valuable. We uncover a potentially surprising connection between the share of preferences where commitment has value and the cardinalities of the action set and the state space (Theorem 3).

Say that preferences are felicitous if Sender can obtain his ideal payoff under cheap talk. If preferences are felicitous, commitment obviously has no value. We show that, for any number of states, the share of felicitous preferences is at least $\frac{1}{|A|^{|A|}}$, where $|A|$ denotes the cardinality of the action set. Hence, in any state space, the share of preferences such that commitment has no value is at least $\frac{1}{|A|^{|A|}}$.

Moreover, as the number of states grows large: (i) the share of felicitous preferences converges to $\frac{1}{|A|^{|A|}}$, and (ii) commitment has no value if and only if preferences are felicitous. Thus, when the state space is large, the share of preferences such that commitment has no value is approximately $\frac{1}{|A|^{|A|}}$. For example, if the action set is binary and there are many states, commitment is valuable for roughly 75% of possible preferences.

We also consider several extensions. First, following Corrao and Dai (2023), we decompose the value of commitment into the value of elicitation and the value of mediation (Myerson 1982; Forges 1986; Salamanca 2021). We analyze how the desire to randomize (under commitment, cheap talk, and mediation) relates to the value of elicitation and mediation. Second, we consider games with infinite state and/or action spaces that are limits of finite games. We show that if the committed Sender does not value randomization in these finite games, then he does not value commitment in the limit. Finally, we explore the relationship between the value of commitment and the value of randomization under commitment when both are strictly positive. In general, there are no restrictions: the value of randomization can be arbitrarily small while the value of commitment is large, and vice versa. Nevertheless, we show that, if the value of randomization is smaller than a particular preference-dependent cutoff, the value of commitment must be smaller than the value of randomization.[3]

[3]In the instances where the value of randomization is arbitrarily small while the value of commitment is large, this cutoff is even smaller than the small value of randomization.

Illustrative example

The workhorse example in the Bayesian-persuasion literature is a prosecutor (Sender) trying to convince a judge (Receiver) to convict a defendant who is guilty or innocent. The judge's preferences are such that she prefers to convict if the probability of guilt is weakly higher than the probability of innocence. The prosecutor has state-independent preferences and always prefers conviction. The prior probability of guilt is 0.3.

If the environment is cheap talk, the unique equilibrium outcome is that the judge ignores the prosecutor and always acquits the defendant. If the prosecutor can commit to an experiment about the state, however, he will conduct a stochastic experiment that indicates guilt whenever the defendant is guilty and indicates guilt with probability $\frac{3}{7}$ when the defendant is innocent (Kamenica and Gentzkow 2011). This experiment induces the judge to convict the defendant with 60% probability. The prosecutor is thus strictly better off than under cheap talk.

Our Theorem 1 tells us that the two facts, (i) the prosecutor's optimal experiment involves randomization and (ii) the prosecutor does better under commitment, imply each other.[4] Of course, the prosecutor-judge example was designed to be extremely simple, so in this particular example one can easily determine the optimal experiment and the value of commitment without our result. In more complicated environments, however, Theorem 1 can simplify the determination of whether commitment is valuable. Except in certain cases, such as uniform-quadratic (Crawford and Sobel 1982) or transparent preferences (Lipnowski and Ravid 2020), cheap-talk games can be difficult to solve. Theorem 1 can then be used to determine whether commitment is valuable without solving for cheap-talk equilibria, simply by computing the Bayesian-persuasion optima and checking whether they include a partitional experiment.[5]

The prosecutor-judge example also illustrates the distinction between the if-and-only-if result in

[4]Theorem 1 only states that (i) and (ii) imply one another for a generic set of preferences. To apply the theorem here, we note that the preferences in the prosecutor-judge example belong to the generic set used in the proof of the theorem. Moreover, in Online Appendix B.2 we show that Theorem 1 holds when Sender has state-independent preferences.

[5]Recent research provides a large toolbox for solving Bayesian-persuasion problems, including concavification (Kamenica and Gentzkow, 2011), price-theoretic approaches (Kolotilin 2018; Dworczak and Martini 2019), duality (Dworczak and Kolotilin 2024), and optimal-transport theory (Kolotilin, Corrao, and Wolitzky 2025). Bergemann and Morris (2016) show that persuasion problems can be formulated as linear programs; it is well known that linear programs can be computed in polynomial time. In contrast, Babichenko et al. (2023) establish that it is NP-hard to approximate Sender's maximum payoff in cheap talk, or even to determine if that payoff is strictly greater than in a babbling equilibrium. For a survey of computational approaches to Bayesian persuasion, see Dughmi (2017).

Theorem 1 and the unidirectional Theorem 2. Recall that Theorem 2 does not claim that the value of commitment is positive only if randomization is valuable in cheap talk. The prosecutor-judge example provides a counterexample to such a claim. In the cheap-talk game, the prosecutor has no value for randomization: with or without it, he never obtains any convictions. Yet, the prosecutor obviously values commitment.

Finally, the prosecutor-judge example also helps illustrate what Theorem 1 does *not* say. Prohibiting randomization would not mean commitment is not valuable. Suppose that the prosecutor is endowed with commitment, but is legally obliged to use only partitional experiments. In that case, the prosecutor would provide a fully informative experiment, obtaining a conviction with 30% probability. That is still better than his cheap-talk payoff of no convictions.

Related literature

Our paper connects the literatures on cheap talk (Crawford and Sobel 1982) and Bayesian persuasion (Kamenica and Gentzkow 2011). Min (2021) and Lipnowski, Ravid, and Shishkin (2022) examine environments with limited commitment that are a mixture of cheap talk and Bayesian persuasion. In contrast, we focus on the question of when cheap talk and Bayesian persuasion yield the same payoff to Sender.[6]

Glazer and Rubinstein (2006) and Sher (2011) consider disclosure games and derive conditions on preferences that imply that Receiver values neither commitment nor randomization. Arieli et al. (2025) establish connections between randomization and efficiency under both cheap talk and Bayesian persuasion. Relatedly, Galperti et al. (2026) show that, in competitive data markets with information intermediaries, equilibria are necessarily inefficient if the intermediaries value randomization.

Several papers examine value of commitment under the assumption that Sender has state-independent preferences. When the action space is finite, as in our framework, Lipnowski and Ravid (2020) show that (for almost every prior) Sender either: (i) obtains his ideal payoff in cheap talk, or (ii) values commitment; Best and Quigley (2024) show that (for almost every prior) Sender either:

[6]Perez-Richet (2014) and Koessler and Skreta (2023) examine the circumstances under which Sender attains his Bayesian-persuasion payoff even if he learns the state prior to selecting the experiment. Goltsman et al. (2009) study the performance of negotiation, mediation, and arbitration relative to cheap talk.

(i) obtains his ideal payoff under the prior, or (ii) values randomization. Titova and Zhang (2025) establish a connection between randomization and the attainability of the Bayesian-persuasion payoff under verifiable messages. Corrao and Dai (2023) examine Sender's payoff under cheap talk, mediation, and Bayesian persuasion. They establish that Sender does not value commitment if his payoffs are the same under mediation and Bayesian persuasion.

In the context of mechanism design, value of commitment and value of randomization have been studied separately. Mechanism design with limited commitment has been studied by Akbarpour and Li (2020) and Doval and Skreta (2022), among others. Value of randomization in mechanism design has been widely recognized in single-agent multi-product monopolist settings (e.g., Manelli and Vincent 2006). In contrast, with two or more agents, Chen et al. (2019) establish that if agents' types are atomless and independently distributed, randomization is never valuable.

2 Set-up and definitions

Preferences and beliefs

Receiver (she) has a utility function $u_R(a, \omega)$ that depends on her action $a \in A$ and the state of the world $\omega \in \Omega$. Both A and Ω are finite; our analysis relies heavily on this assumption.[7] For any finite set X, we denote its cardinality by $|X|$. Sender (he) has a utility function $u_S(a, \omega)$ that depends on Receiver's action and the state. The players share an interior common prior μ_0 on Ω. We say action a^* is i's *ideal action in* ω if $a^* \in \arg\max_{a \in A} u_i(a, \omega)$.

Environments, shares, and genericity

We refer to the pair (u_S, u_R) as the (preference) *environment*.

Since u_S and u_R have a finite domain, they are bounded. We further restrict our attention to environments where u_S and u_R take values in some fixed interval, which, without loss of generality, we set to $[0, 1]$. Under these assumptions, the set of all environments is isomorphic to $[0, 1]^{2|A||\Omega|}$. When we say that a claim holds for a γ share of environments, we simply mean that the set of

[7] At the risk of being excessively philosophical, we consider environments with finite A and Ω to be more realistic; the use of infinite sets often provides tractability but rarely improves realism. We discuss the role of the finiteness assumption in Section 7.2 and Online Appendix B.1.

environments where the claim holds has Lebesgue measure γ on $\mathbb{R}^{2|A||\Omega|}$.

We say a set of environments is *generic* if it has Lebesgue measure one on $\mathbb{R}^{2|A||\Omega|}$.[8] When we say that a claim holds *generically*, we mean that it holds for a generic set of environments.[9] When we say that, given A, a claim holds *generically as* $|\Omega| \to \infty$, we mean that the share of environments where the claim does not hold converges to zero as $|\Omega| \to \infty$. We say a claim is *not-generically-true* if there exists a strictly positive share of environments where the claim does not hold.

Cheap talk, Bayesian persuasion, and value of commitment

Let M be a finite message space with $|M| > \max\{|\Omega|, |A|\}$.[10] Sender chooses a messaging strategy $\sigma : \Omega \to \Delta M$. Receiver chooses an action strategy $\rho : M \to \Delta A$.

A profile of strategies (σ, ρ) induces expected payoffs

$$U_i(\sigma, \rho) = \sum_{\omega,m,a} \mu_0(\omega)\, \sigma(m|\omega)\, \rho(a|m)\, u_i(a, \omega) \quad \text{for } i = S, R.$$

A profile (σ^*, ρ^*) is *S-BR* if $\sigma^* \in \arg\max_\sigma U_S(\sigma, \rho^*)$. A profile (σ^*, ρ^*) is *R-BR* if $\rho^* \in \arg\max_\rho U_R(\sigma^*, \rho)$.

Sender's *ideal payoff* is the maximum U_S induced by any profile.

A *cheap-talk equilibrium* is a profile that satisfies S-BR and R-BR.[11] We define (Sender's) *cheap-talk payoff* as the maximum U_S induced by a cheap-talk equilibrium.[12]

A *persuasion profile* is a profile that satisfies R-BR. The (Bayesian) *persuasion payoff* is the

[8] Our results also hold if we use a topological rather than a measure-theoretic notion of genericity. See Footnotes 35 and 37 in the Appendix.

[9] Lipnowski (2020), who focuses on finite action and state spaces as we do, establishes that commitment has no value when Sender's value function over Receiver's beliefs is continuous. Such continuity, however, holds for a zero share of environments if Receiver's optimal action depends on the state. In contrast, we focus on results that hold generically.

[10] Our results concern Sender's payoffs under cheap talk, Bayesian persuasion, and restriction to partitional strategies in those models. To derive Sender's maximal payoff, it is without loss of generality to set $|M| \geq |\Omega|$ for cheap talk (Matthews 1990), $|M| \geq \min\{|\Omega|, |A|\}$ for Bayesian persuasion (Kamenica and Gentzkow 2011), and $|M| \geq |\Omega|$ for partitional strategies (trivially). Therefore, assuming $|M| \geq |\Omega|$ would suffice for our results. However, further assuming $|M| \geq |A| + 1$ simplifies the proofs of Lemmas 4 and 8.

[11] This definition may seem unconventional since it uses Nash equilibrium, rather than perfect Bayesian equilibrium, as the solution concept. In cheap-talk games, however, the set of equilibrium outcomes (joint distributions of states, messages, and actions) is exactly the same whether we apply Nash or perfect Bayesian as the equilibrium concept. The formulation in terms of Nash equilibria streamlines the proofs.

[12] Throughout, we examine the value of commitment to Sender; hence the focus on Sender's payoff. The set of equilibrium payoffs is compact so a maximum exists.

maximum U_S induced by a persuasion profile.[13] We refer to a persuasion profile that yields the persuasion payoff as *optimal*.

The value of commitment is the difference between the persuasion payoff and the cheap-talk payoff. If this value is strictly positive, we say *commitment is valuable*; otherwise, we say *commitment has no value*.

Note that our definition of the value of commitment compares the persuasion payoff to the *best* payoff achievable under cheap talk. For example, if $u_S = u_R$, we would say that commitment has no value since full revelation is a cheap-talk equilibrium. Nevertheless, even in such circumstances, investing in commitment might be worthwhile since a committed Sender avoids the risk of the babbling equilibrium. In other words, throughout, we focus on the value that commitment provides over and above its benefit in equilibrium selection.[14]

Partitional strategies and value of randomization

A messaging strategy σ is *partitional* if for every ω, there is a message m such that $\sigma(m|\omega) = 1$. A profile (σ, ρ) is a *partitional profile* if σ is partitional.[15] The *partitional persuasion payoff* is the maximum U_S induced by a partitional persuasion profile. The *partitional cheap-talk payoff* is the maximum U_S induced by a partitional cheap-talk equilibrium.[16]

Value of randomization under commitment is the difference between the persuasion payoff and the partitional persuasion payoff. If this value is strictly positive, we say that *committed Sender values randomization*. Analogously, *value of randomization under cheap talk* is the difference between the cheap-talk payoff and the partitional cheap-talk payoff. If this value is strictly positive, we say that *cheap-talk Sender values randomization*.

[13] Lipnowski, Ravid, and Shishkin (2024) establish that, with finite A and Ω, this is generically the only payoff that Sender could attain in an equilibrium of a Bayesian-persuasion game.

[14] Our results do not concern the equilibrium-selection benefit, but we acknowledge that this benefit is important. Whenever committed Sender wishes to provide information, commitment provides the equilibrium-selection benefit as every cheap-talk game admits a babbling equilibrium. And even if no information is the commitment optimum, there still might be an equilibrium-selection benefit since there might be cheap-talk equilibria that yield a payoff to Sender that is lower than babbling.

[15] Our focus is on the connection between Sender's value of commitment and Sender's randomization. Consequently, the definition of a partitional profile only concerns Sender's strategy. That said, along the way we will establish a result about Receiver playing pure strategies (see Lemma 4).

[16] A partitional cheap-talk equilibrium always exists because the babbling equilibrium outcome can be supported by Sender always sending the same message.

3 Value of commitment and randomization under commitment

In this section, we consider a Sender with commitment power, who can choose his messaging strategy prior to being informed of the state. We ask whether this commitment power makes Sender strictly better off. We link the value of commitment to Sender's behavior under commitment, in particular to whether Sender has a strict preference for randomization.

Theorem 1. *Generically, commitment is valuable if and only if committed Sender values randomization.*

Here we provide an intuition about the only-if direction of the theorem. We postpone the discussion of the converse until the next section, as the intuition for it is related to that for Theorem 2. Formal proofs are in the Appendix.[17]

Let μ_E denote the posterior belief induced by learning that ω is in $E \subseteq \Omega$. For a generic set of environments, Receiver's optimal action given any $\mu \in \{\mu_E\}_{E \subseteq \Omega}$ is unique and remains optimal in a neighborhood of beliefs around μ.

Now, suppose that there is a partitional optimal persuasion profile (σ, ρ). Let M_σ be the set of messages that are sent under σ. Because σ is partitional, each $m \in M_\sigma$ is associated with a subset of the state space, namely $\Omega_m := \{\omega | \sigma(m|\omega) = 1\}$. For each $m \in M_\sigma$, let μ_m be the belief induced by m, and let a_m be Receiver's (uniquely) optimal action given μ_m. As noted above, a_m remains optimal in a neighborhood of beliefs around μ_m.

Key to the proof is to note that every action a_m taken in equilibrium must be Sender's preferred action, among the actions taken in equilibrium, in all states where action a_m is taken. In other words, let $A^* = \{a_m | m \in M_\sigma\}$; for every $m \in M_\sigma$, for every $\omega \in \Omega_m$, we have $u_S(a_m, \omega) \geq u_S(a_{m'}, \omega)$ for all $a_{m'} \in A^*$. Why does this hold? If it were not the case, Sender could attain a higher payoff with an alternative strategy: if $u_S(a_m, \omega) < u_S(a_{m'}, \omega)$ for some $a_{m'} \in A^*$, $\omega \in \Omega_m$, Sender could send m' in ω with a small probability and still keep a_m optimal given m.

Finally, the fact that for every $m \in M_\sigma$, $u_S(a_m, \omega) \geq u_S(a_{m'}, \omega)$ for all $a_{m'} \in A^*$ and all $\omega \in \Omega_m$ implies that (σ, ρ) is a cheap-talk equilibrium.[18] Hence, commitment is not valuable.

[17] Theorem 1 can be extended to establish a threefold equivalence. Generically, the following imply each other: (i) commitment is valuable, (ii) committed Sender values randomization, and (iii) any optimal persuasion profile induces a belief under which Receiver has multiple optimal actions (see Theorem $1'$ in the Appendix).

[18] Deviating to an on-path message $\hat{m} \in M_\sigma$ cannot be profitable by the inequality $u_S(a_m, \omega) \geq u_S(a_{\hat{m}}, \omega)$ for

Before we turn to the next section, it is perhaps worth noting that Theorem 1 implies that – in finite worlds – whenever committed Sender wishes to fully reveal the state, full revelation is also a cheap-talk equilibrium. Of course, this is not true in the canonical quadratic environment in Crawford and Sobel (1982) with $A = \Omega = [0, 1]$. In Online Appendix B.1.1, we illustrate the role of finiteness in this environment.

4 Value of commitment and randomization in cheap talk

In this section, we consider a Sender without commitment power who engages in a cheap-talk game. We ask whether he would be strictly better off if he had commitment power. We link the value of such commitment to Sender's behavior in Sender-preferred cheap-talk equilibria, in particular to whether Sender necessarily randomizes in such equilibria.

Theorem 2. *Generically, commitment is valuable if cheap-talk Sender values randomization.*

Theorem 2 and the if-direction of Theorem 1 both derive from the following result. Generically, if a cheap-talk equilibrium yields the persuasion payoff, then there is a partitional σ and a (pure-strategy) ρ such that (σ, ρ) is a cheap-talk equilibrium and yields the persuasion payoff. We build this result (Proposition 3 in Appendix A.3) in two steps.

The first step (Lemma 4) shows that, generically, if (σ, ρ) is R-BR and yields the persuasion payoff, then ρ must be pure-on-path. Consider toward contradiction that there is a message m sent with positive probability under σ, and there are two distinct actions, say a and a', in the support of $\rho(\cdot|m)$. It must be that both Sender and Receiver are indifferent between a and a' under μ_m, the belief induced by m: Receiver has to be indifferent because (σ, ρ) is R-BR; Sender has to be indifferent because (σ, ρ) yields the persuasion payoff, which maximizes U_S over all persuasion profiles.[19] The result then follows from establishing that such a coincidence of indifferences generically cannot arise when Sender is optimizing. For some intuition for why this is the case, consider Figure 1 which illustrates this result when there are three states. Suppose a_1 and a_2 are in the support of $\rho(\cdot|m)$. Region R_i denotes beliefs where Receiver prefers a_i. Region S_i denotes beliefs where

$\hat{m} \in M_\sigma$; for any off-path message $\hat{m} \notin M_\sigma$, we can just set $\rho(\cdot|\hat{m}) = \rho(\cdot|m^*)$ for some $m^* \in M_\sigma$, thus ensuring that such a deviation is also not profitable.

[19] If Sender strictly prefers one action over the other, say a over a', at μ_m, then Sender would obtain a higher payoff if Receiver always takes a following m (which would remain R-BR given Receiver's indifference).

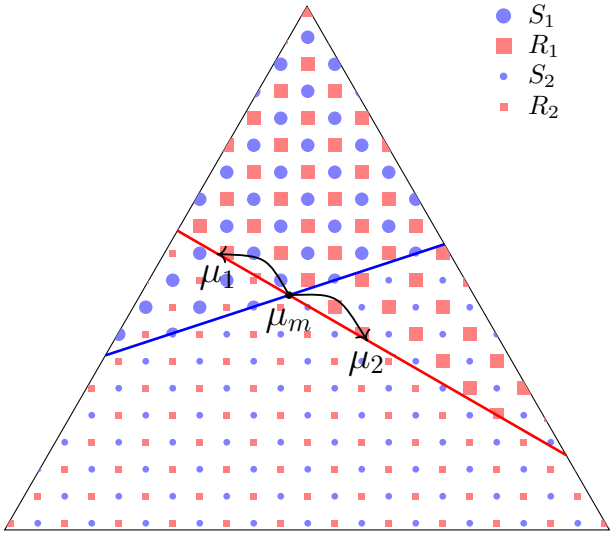


Figure 1: Indifference incompatible with optimality

Sender prefers a_i. Generically, the border between R_1 and R_2 is distinct from the border between S_1 and S_2 and thus the two borders have at most one intersection, μ_m. Moreover, generically μ_m (if it exists) is an interior belief. But now, Sender could deviate to an alternate strategy that induces beliefs μ_1 and μ_2 instead of μ_m, with Receiver still indifferent between a_1 and a_2 at both μ_1 and μ_2. Suppose that Receiver takes action a_i following belief μ_i. This strategy is still R-BR for Receiver and gives Sender a strictly higher payoff. Thus, we have reached a contradiction. With more than three states and more than two actions, the proof that the coincidence of indifferences generically cannot arise is conceptually similar but notationally more involved. It is presented in the Appendix as Lemma 2.

The second step (Lemma 5) shows that, generically, if (σ, ρ) is a cheap-talk equilibrium that yields the persuasion payoff, and ρ is a pure strategy on-path, then there is a partitional cheap-talk equilibrium that yields the persuasion payoff. This is easy to see. Generically, for any ω and any $a \neq a'$, we have $u_S(a, \omega) \neq u_S(a', \omega)$. Now, consider some cheap-talk equilibrium (σ, ρ), with ρ pure-on-path, that yields the persuasion payoff. If σ is partitional, our result is immediate. Suppose to the contrary that in some ω, both m and m' are sent with positive probability. Then, m and m' must induce the same action: if m induces some a and m' induces a distinct a', the fact that $u_S(a, \omega) \neq u_S(a', \omega)$ would mean that σ cannot be S-BR. Given that any two messages sent in ω induce the same action, we can define $\rho(\sigma(\omega))$ as the action that Receiver takes in state ω given

(σ, ρ).

Now, we can consider an alternative, partitional profile $(\hat{\sigma}, \hat{\rho})$. Let f be any injective function from A to M. Let $\hat{\sigma}(\omega) = f(\rho(\sigma(\omega)))$ and $\hat{\rho}(f(a)) = a$. It is immediate that $(\hat{\sigma}, \hat{\rho})$ is also a cheap-talk equilibrium and yields the persuasion payoff.

It is perhaps worth noting that Theorems 1 and 2 jointly imply the following:

Corollary 1. *Generically, if cheap-talk Sender values randomization, then committed Sender values randomization.*

5 Application to grading

For an application of our results, we consider their implications for grading policies. This application also clarifies a sense in which "randomization" in the statement of our results need not be interpreted literally.[20]

Suppose Sender is a school that assigns grades to its students. We interpret M as the set of potential grades. Each student is characterized by a vector of attributes. We say an attribute is *relevant* if an employer values it or the school's value of placing the student with an employer depends on it. We interpret Ω as the set of all possible configurations of the relevant attributes. We maintain the assumption that Ω is finite.

Students also have irrelevant attributes. We denote by X the set of all possible configurations of the irrelevant attributes. We assume that the distribution over X is atomless. The school utilizes a deterministic *grading scheme* $g : \Omega \times X \to M$. We say a grading scheme g is *fair* if $g(\omega, x) = g(\omega, x')$ for every ω, x, x'. Otherwise, the scheme is *unfair*.[21]

For this application, instead of envisioning a single Receiver, we assume that each student

[20] A large literature examines the economics of grading policies. Ostrovsky and Schwarz (2010) consider equilibrium dynamics when multiple schools, whose students all compete for the same jobs, select grading policies. Boleslavsky and Cotton (2015) analyze how the grading policy equilibrium impacts the school's incentives to invest in quality. Dubey and Geanakoplos (2010) examine the impact of grading policies on student motivation when students care about their relative rank. Boleslavsky and Kim (2018) consider the problem faced by a school whose grading policy simultaneously impacts students' effort and employers' willingness to hire a student with a specific grade.

[21] One may view our notion of fairness as being too permissive. Even a grading scheme that satisfies $g(\omega, x) = g(\omega, x')$ for every ω, x, x' may be seen as unfair: the set of attributes deemed relevant may include the likelihood that the student will become a donor, or grades may provide insufficient information (e.g., everyone gets an A), etc. Alternatively, therefore, we could postulate only that a grading scheme is unfair if $g(\omega, x) \neq g(\omega, x')$ for some ω, x, x', without taking a stance on which schemes are fair. Then, Theorem 1 would imply that, generically, any school that values commitment prefers to grade unfairly, though the converse need not hold.

applies to a distinct employer. Each employer observes the grade $m \in M$ of its applicant and chooses one of finitely many actions $a \in A$ (e.g., whether to hire the student and if so for what position). All employers have the same utility function $u_R(a, \omega)$ that depends on the employer's action and the relevant attributes of the applicant. (If there were a single employer who observed the grades of all of the applicants, this would effectively provide Sender with some commitment power because the distribution of messages would be directly observable to Receiver.) The school's utility is additive across its students; for each student, the school's payoff $u_S(a, \omega)$ depends on that student's outcome and that student's relevant attributes.

Under *discretionary grading*, the school freely chooses a grade to assign to each student, i.e., the school selects any grading scheme it wishes. The employer only observes its applicant's grade but not the grading scheme that was used. Alternatively, the school could implement a publicly observable *grading policy* that restricts the set of schemes that it can use.

There are active debates about whether colleges in the United States should employ grading policies, especially given the context of grade inflation (Ostrovsky and Schwarz 2010; Gans and Kominers 2026). A grading policy could be a restriction to a specific grading scheme; this would make the situation equivalent to Bayesian persuasion.[22] Another type of grading policy is one where the school commits to a given distribution of grades (Lin and Liu 2024). We refer to such a policy as a *mandated curve*. For example, the University of Chicago Law School mandates a pre-specified share of students that will receive a given narrow range of numerical grades. Another form of commitment is a GPA cap. For example, the University of Chicago Booth School of Business mandates that the average grade assigned in a given course must not exceed B+. Yet another type of policy restricts the share of students who may receive a given grade. From 2004 until 2014, Princeton University mandated that no more than 35 percent of students should receive an A in an undergraduate class. In May of 2026, the Harvard Faculty of Arts and Sciences voted to limit the number of A grades in an undergraduate class to 20 percent plus an additional four A's, starting in the fall of 2027.

We say that *the school values commitment* if it strictly prefers to implement any grading policy

[22]By conditioning the grade on the irrelevant attributes, the school can implement any distribution of grades conditional on each ω. The formulation of experiments as deterministic functions of an expanded state space was introduced by Gentzkow and Kamenica (2017) and Green and Stokey (2022). It has been further studied in Brooks, Frankel, and Kamenica (2022) and Brooks, Frankel, and Kamenica (2024).

(full commitment, mandated curve, GPA cap, limited share of A's, etc.) over discretionary grading. We know that any policy must yield a payoff that is weakly lower than full commitment, so if a grading policy yields a strictly higher payoff than discretionary grades, we know that the persuasion payoff (full commitment) exceeds the cheap-talk payoff (discretionary grades).

We say that *the school prefers to grade unfairly* if its optimal grading scheme is unfair. In other words, if the school were able to commit to any grading scheme, it would commit to an unfair one.

Theorem 1 tells us that, generically, the school values commitment if and only if it prefers to grade unfairly.[23] Thus, whenever we observe a school mandating a curve or a GPA cap, we know that the school's ideal policy is unfair.[24]

6 How often is commitment valuable?

Theorems 1 and 2 by themselves do not shed light on the types of environments where commitment (and thus randomization) is valuable nor about how common such environments are. This section aims to fill that gap.

Say that the environment is *felicitous* if Sender obtains his ideal payoff under cheap talk.[25] It is immediate that commitment has no value if the environment is felicitous. The converse, of course, is not generally true (e.g., $u_S = -u_R$).

We first show that, for any Ω, the share of environments that are felicitous is at least $\frac{1}{|A|^{|A|}}$. Thus, in general, the share of environments such that commitment has no value must be greater than $\frac{1}{|A|^{|A|}}$.

We then establish that, in the limit as Ω grows large: $(*)$ the share of environments that are felicitous converges to exactly $\frac{1}{|A|^{|A|}}$, and $(**)$ generically, commitment has no value *if and only if* the environment is felicitous, i.e., the non-felicitous environments where commitment has no value (such as $u_S = -u_R$) become vanishingly rare. Thus, as $|\Omega| \to \infty$, the share of environments such

[23] Even if we observe a school mandating a curve, Theorem 1 does not imply that the school will implement an unfair scheme if it can only commit to a mandated curve (i.e., is unable to fully commit to a particular scheme).

[24] Our analysis views the school (that cares about student placements) and the professor (who is assigning grades) as a single agent. A distinct motivation for a grading policy such as a GPA cap, outside of our Sender-Receiver framework, is an agency conflict between the school and the professor. For example, the professor may wish to give uniformly high grades in order to avoid student complaints so the school might impose a GPA cap to mitigate that temptation (Frankel 2014). Moreover, a grading policy could have a distinct benefit of aiding equilibrium coordination about the meaning of grades; our focus on Sender-preferred equilibria assumes miscoordination away.

[25] We thank Roger Myerson for suggesting this term.

that commitment has no value converges exactly to $\frac{1}{|A|^{|A|}}$.

Theorem 3. *Fix A.*

(i) For any Ω, the share of environments such that commitment has no value is at least $\frac{1}{|A|^{|A|}}$.

(ii) As $|\Omega| \to \infty$,

(a) generically, commitment has no value if and only if Sender obtains his ideal payoff under cheap talk.

(b) the share of environments such that commitment has no value converges to $\frac{1}{|A|^{|A|}}$.

Here we provide a sketch of the proof. For each $a \in A$, let Ω_a be the set of states where a is Sender's ideal action. Generically, if $a \neq a'$, Ω_a and $\Omega_{a'}$ do not intersect; consider this case. Then, it is easy to see that the environment is felicitous if and only if for each $a, a' \in A$, we have[26]

$$\sum_{\omega \in \Omega_a} \mu_0(\omega) \left(u_R(a, \omega) - u_R\left(a', \omega\right) \right) \geq 0. \tag{1}$$

Now, for any Ω_a that is not empty, the share of Receiver's preferences on $A \times \Omega_a$ such that inequality (1) is satisfied is $\frac{1}{|A|}$. Thus if all of Ω_a's are non-empty, the share of environments that are felicitous is $\left(\frac{1}{|A|}\right)^{|A|}$, or $\frac{1}{|A|^{|A|}}$.

If an Ω_a is empty, inequality (1) is satisfied vacuously for that a. Thus, the share of felicitous environments is weakly greater than $\frac{1}{|A|^{|A|}}$. Since commitment has no value in felicitous environments, we conclude that commitment has no value in at least $\frac{1}{|A|^{|A|}}$ share of environments.

We establish part (ii) of the theorem by showing that as $|\Omega|$ grows large: $(*)$ the share of preferences such that an Ω_a is empty converges to zero so the share of environments that are felicitous converges to $\frac{1}{|A|^{|A|}}$, and $(**)$ the share of environments such that commitment has no value converges to the share of environments that are felicitous.

Part $(*)$ is easy to see. For any $a \in A$, as Ω grows large, the share of preferences such that there is *no* state where a is Sender's ideal action converges to zero.

[26]This condition also appears in Antic, Chakraborty, and Harbaugh (2025) and Aybas and Callander (2024). In Antic, Chakraborty, and Harbaugh (2025), it is a necessary condition for the possibility of subversive conversations: without it, a third-party with veto power would prevent a committee from implementing a project solely based on the information that the committee wants to do so. Aybas and Callander (2024) consider preferences of the form $u_R(a, \omega(\cdot)) = \omega(a)^2$ and $u_S(a, \omega(\cdot)) = (\omega(a) - b)^2$ for some $b > 0$ where $\omega : A \to \mathbb{R}$ is the realized path of a Brownian motion. They identify features of b and A such that the condition holds.

To establish part $(**)$, say that an environment is *jointly-inclusive* if for every action a, there is some state ω such that a is an ideal action for both Sender and Receiver in ω. Analogously to part $(*)$, it is easy to see that as Ω grows large, the share of environments that are jointly-inclusive converges to 1. To complete the proof of part $(**)$, we argue that, generically, if the environment is jointly-inclusive and commitment has no value, then the environment must be felicitous. First, because commitment has no value, Theorem 2 implies that there is a partitional profile (σ, ρ) that is a cheap-talk equilibrium and yields the persuasion payoff. Next, we note that every action $a \in A$ must be induced by (σ, ρ): there is a state ω where a is both Sender's and Receiver's ideal action, so if a were never taken, the committed Sender could profitably deviate by sometimes revealing ω and inducing a,[27] thus contradicting the fact that (σ, ρ) yields the persuasion payoff. This in turn implies that, for every ω, $\rho(\sigma(\omega))$ must be Sender's ideal action in ω. (If Sender strictly preferred some other a' in ω, (σ, ρ) could not be S-BR as the cheap-talk Sender would profitably deviate and set $\sigma(\omega)$ to be whatever message induces a'; since all actions are induced by (σ, ρ), there must be such a message.) Thus, (σ, ρ) is a partitional profile that is R-BR and induces Receiver to take Sender's ideal action in every state. But this means that every message sent under σ fully reveals what action is ideal for Sender, and Receiver complies and takes that action. Hence, the environment is felicitous.

We conclude this section with a few comments about Theorem 3.

First, whether commitment has value in a given environment (u_S, u_R) depends on the prior μ_0. Yet, Theorem 3 remarkably holds for any (interior) prior.

Second, a corollary of Theorem 3, part (ii)(b), is that in the limit $\lim_{|A|\to\infty} \lim_{|\Omega|\to\infty}$, commitment generically has value. One might be tempted to summarize this corollary as "if both A and Ω are large, then commitment is almost certainly valuable." An issue with this summary, however, is that, at least given our current proof method, the conclusion relies on the order of limits. Whether commitment is generically valuable when the order of the limits is reversed (i.e., $\lim_{|\Omega|\to\infty} \lim_{|A|\to\infty}$) or we consider the joint limit (i.e., $\lim_{\substack{|\Omega|\to\infty \\ |A|\to\infty}}$) remain open questions.

Third, the felicity condition has a flavor of alignment of Sender's and Receiver's preferences. However, it does not preclude the possibility that Receiver is much worse off than she would be if

[27] Sender could reveal ω with some probability ε; Receiver's response to all other messages would remain unchanged if ε is sufficiently small.

Sender's and Receiver's preferences were fully aligned. For instance, consider the prosecutor-judge example and suppose that the prior is 0.7 rather than 0.3; then, the environment is felicitous but Receiver obtains no information.

7 Discussion

7.1 Mediation

Our analysis so far compares Bayesian persuasion and cheap talk. A third benchmark, mediated communication, sits between the two (Myerson 1982; Forges 1986; Salamanca 2021; Corrao and Dai 2023). Under mediation, Sender freely sends a message to a trusted device that then generates (through a commonly understood protocol) a possibly random message to Receiver.

Formally, a *mediation device* is a mapping $d : M \to \Delta M$. Sender chooses a strategy $\sigma : \Omega \to \Delta M$ and Receiver chooses a strategy $\rho : M \to \Delta A$. Since the message sent by Sender need not coincide with the message observed by Receiver, we use m_S for the former and m_R for the latter. A device d and a profile (σ, ρ) induce expected payoffs

$$U_i(\sigma, \rho; d) = \sum_{\omega, m_S, m_R, a} \mu_0(\omega)\sigma(m_S|\omega)d(m_R|m_S)\rho(a|m_R)u_i(a, \omega) \quad \text{for } i = S, R.$$

A profile (σ^*, ρ^*) is a *mediated equilibrium given* d if $\sigma^* \in \arg\max_\sigma U_S(\sigma, \rho^*; d)$ and $\rho^* \in \arg\max_\rho U_R(\sigma^*, \rho; d)$. A profile (σ^*, ρ^*) is a *mediated equilibrium* if it is a mediated equilibrium given d for some d.

The *mediation payoff* is the maximum U_S induced by a mediated equilibrium. A device is *deterministic* if for every m_S there exists a m_R such that $d(m_R|m_S) = 1$. The *deterministic mediation payoff* is the maximum U_S induced by a mediated equilibrium given a deterministic device. The *value of randomization under mediation* is the difference between the mediation payoff and the deterministic mediation payoff.

Let V^{CT}, V^{MD}, and V^{BP} denote the cheap-talk, mediation, and persuasion payoffs. Trivially, $V^{CT} \leq V^{MD} \leq V^{BP}$.[28]

[28] A cheap-talk equilibrium is a mediated equilibrium given the identity device. Given any mediated equilibrium (σ^*, ρ^*) given d, Sender with commitment can employ the strategy $\sigma(m|\omega) = \sum_{m_S} d(m|m_S)\sigma^*(m_S|\omega)$ and achieve the same payoff.

Following Corrao and Dai (2023), we can therefore decompose the value of commitment (CV) into the value of elicitation (EV) and the value of mediation (MV):

$$CV := V^{BP} - V^{CT}, \qquad EV := V^{BP} - V^{MD}, \qquad MV := V^{MD} - V^{CT}.$$

The value of mediation is the gain from routing communication through a device that can garble Sender's reports; the value of elicitation is the further gain from conditioning the experiment on the state directly, rather than having to elicit the state from Sender. Clearly, $CV = EV + MV$.

Our main results tie the value of commitment to Sender's desire to randomize. How do the two components of that value relate to the desire to randomize under each benchmark? Let $\Delta_{\mathcal{R}}^{BP}$, $\Delta_{\mathcal{R}}^{MD}$, and $\Delta_{\mathcal{R}}^{CT}$ denote the values of randomization under commitment, mediation, and cheap talk, respectively.[29] Recall that a claim is not-generically-true if there exists a strictly positive share of environments where the claim does not hold.

Proposition 1.

(i) $\Delta_{\mathcal{R}}^{MD} > 0 \Leftrightarrow MV > 0$.

(ii) Generically, $EV > 0 \Rightarrow \Delta_{\mathcal{R}}^{BP} > 0$.

(iii) It is not-generically-true that $\Delta_{\mathcal{R}}^{BP} > 0 \Rightarrow EV > 0$.

(iv) It is not-generically-true that $\Delta_{\mathcal{R}}^{MD} > 0 \Rightarrow EV > 0$.

(v) It is not-generically-true that $EV > 0 \Rightarrow \Delta_{\mathcal{R}}^{MD} > 0$.

(vi) It is not-generically-true that $\Delta_{\mathcal{R}}^{CT} > 0 \Rightarrow MV > 0$.

(vii) It is not-generically-true that $MV > 0 \Rightarrow \Delta_{\mathcal{R}}^{CT} > 0$.

The intuition behind part (i) is very simple: the set of Sender's payoffs that are achievable with a deterministic mediation device is the same as the set of Sender's payoffs that are achievable in cheap talk.[30]

[29] There are, in principle, 18 pairwise comparisons (e.g., $\Delta_{\mathcal{R}}^{BP} > 0 \Rightarrow CV > 0$) between $\Delta_{\mathcal{R}}^{BP}$, $\Delta_{\mathcal{R}}^{MD}$, and $\Delta_{\mathcal{R}}^{CT}$ on one hand, and CV, EV, and MV on the other. We examine the eight natural ones, in addition to the four analyzed earlier. Asking, say, whether $\Delta_{\mathcal{R}}^{BP} > 0$ implies $MV > 0$ would be strained: MV compares mediated communication with cheap talk, a comparison in which Bayesian persuasion plays no part.

[30] Corollary 2 of Corrao and Dai (2023) establishes a similar result.

Part (ii) follows from Theorem 1. Since $CV = EV + MV$ and both components are non-negative, a strictly positive value of elicitation implies a strictly positive value of commitment.

For parts (iii) and (iv), we construct a positive share of environments where $V^{BP} = V^{MD} > V^{CT}$ with V^{CT} equal to the babbling payoff. In such environments, $CV = MV > 0$ and $EV = 0$. Theorem 1 implies that $\Delta_{\mathcal{R}}^{BP} > 0$, which establishes part (iii). Part (i) implies that $\Delta_{\mathcal{R}}^{MD} > 0$, which establishes (iv).

For part (v), recall the prosecutor-judge example. Conviction strictly dominates for the prosecutor, so truth-telling implies that every report he submits to the device must yield the same probability of conviction.[31] Thus, mediation does no better than babbling. Hence $MV = 0$ and, by part (i), the mediator does not value randomization. The prosecutor nonetheless values commitment, and since $MV = 0$, the entire value of commitment stems from elicitation. The same logic applies to a positive share of environments in a neighborhood of this example.

For part (vi), we construct a positive share of environments where $V^{MD} = V^{CT}$ and cheap-talk Sender values randomization. For part (vii), we return to the environments used to prove parts (iii) and (iv): mediation is valuable, but the best cheap-talk equilibrium is babbling, which is trivially attainable with a partitional strategy.

Table 1 summarizes Proposition 1, and Theorems 1 and 2.

	$CV > 0$	$EV > 0$	$MV > 0$
$\Delta_{\mathcal{R}}^{BP} > 0$	$\Leftrightarrow$	$\not\Rightarrow$ $\Leftarrow$	$-$
$\Delta_{\mathcal{R}}^{MD} > 0$	$-$	$\not\Rightarrow$ $\not\Leftarrow$	$\Leftrightarrow$
$\Delta_{\mathcal{R}}^{CT} > 0$	$\Rightarrow$ $\not\Leftarrow$	$-$	$\not\Rightarrow$ $\not\Leftarrow$

Table 1: Relationships between values of randomization and CV, EV, MV.

[31] By the revelation principle (Myerson 1982; Forges 1986), to characterize the feasible state-action distributions under mediation, it suffices to consider direct devices that elicit the state and recommend actions. In other words, a state-action distribution $\pi : \Omega \to \Delta A$ can be induced by a mediated equilibrium if and only if π satisfies (i) for all $\omega, \omega' \in \Omega$, $\sum_{a \in A} [\pi(a|\omega) - \pi(a|\omega')] u_S(a, \omega) \geq 0$, and (ii) for all $a, a' \in A$, $\sum_{\omega} \mu_0(\omega) \pi(a|\omega) [u_R(a, \omega) - u_R(a', \omega)] \geq 0$.

7.2 Infinite Ω or A

Our main results are stated for games with finite Ω and A. In Online Appendix B.1, we explain the role of the finiteness assumption and develop an extension of our results for infinite games by viewing them as limits of finite games.

When the prior is atomless (so the state space is uncountable), the need for randomization can typically be eliminated through purification (Corollary 1 in Zeng 2023) or partitional approximation (Lemma 2 in Ali, Kleiner, and Zhang 2026) under certain regularity conditions. Therefore, it is not useful to examine the implications of the (always absent) value of randomization.

To illustrate the importance of the finite action space, we consider a modification of the canonical Crawford and Sobel (1982) uniform-quadratic example. In particular, suppose that $\Omega = \{0, 1\}$ with $u_R(a, \omega) = -(a - \omega)^2$ and $u_S(a, \omega) = -(a - \omega - b)^2$ for some $b > \frac{1}{2}$. If A were $[0, 1]$ – an uncountable set – the committed Sender would wish to fully reveal the state but could not do so in any cheap-talk equilibrium. Hence, in "violation" of Theorem 1, commitment has value even though committed Sender does not value randomization. By contrast, if we consider any finite discretization of the unit interval, $A = \left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1\right\}$, Theorem 1 applies. The fact that commitment is valuable implies that committed Sender values randomization.[32]

Despite the fact that our results do not directly apply to infinite games, we can leverage some of our results to derive value of commitment in certain infinite games, formulated as limits of generic finite games. In particular, consider a game Γ with a convex action space and with Receiver's payoff strictly concave in the action. If Γ is a limit of a sequence of generic, finite games $\{\mathcal{G}^n\}$ and randomization is not valuable in each $\mathcal{G}^n$, then commitment has no value in Γ. We develop this result formally in Online Appendix B.1.2.

7.3 Away from zeroes

Theorem 1 only tells us that, generically, commitment has zero value if and only if randomization has zero value. A natural question is whether, generically, small value of commitment implies or is implied by small value of randomization. The answer, in general, is no. In Online Appendix B.3, we construct a positive measure of environments where the value of randomization is arbitrarily

[32] We develop this observation fully in Online Appendix B.1.1.

small but the value of commitment is not, and a positive measure of environments where the value of commitment is arbitrarily small but the value of randomization is not.

Nevertheless, there is an environment-specific cutoff such that, if the value of randomization is strictly smaller than this cutoff, the value of commitment is smaller than the value of randomization. Fix some environment (u_S, u_R). Let $\delta := \min_{a \neq a', \omega} |u_S(a, \omega) - u_S(a', \omega)|$ denote the smallest difference in Sender's payoff across all action pairs in any state. For any non-empty $\hat{\Omega} \subseteq \Omega$, let $A^R(\hat{\Omega}) := \arg\max_{a \in A} \sum_{\omega \in \hat{\Omega}} \mu_0(\omega) u_R(a, \omega)$ and $A^*(\hat{\Omega}) := \arg\max_{a \in A^R(\hat{\Omega})} \sum_{\omega \in \hat{\Omega}} \mu_0(\omega) u_S(a, \omega)$. In other words, $A^*(\hat{\Omega})$ is the set of Sender-preferred Receiver-optimal actions when the state is revealed to be in $\hat{\Omega}$. Let $a^*\left(\hat{\Omega}\right)$ denote an arbitrary element of $A^*(\hat{\Omega})$. Let

$$\kappa(\hat{\Omega}) := \frac{\sum_{\omega \in \hat{\Omega}} \mu_0(\omega) u_R(a^*\left(\hat{\Omega}\right), \omega) - \max_{a' \in A \setminus \{a^*(\hat{\Omega})\}} \sum_{\omega \in \hat{\Omega}} \mu_0(\omega) u_R(a', \omega)}{\max_{a, a', \omega} |u_R(a, \omega) - u_R(a', \omega)|}$$

be the minimum slackness of the obedience constraint given $\hat{\Omega}$.[33] Finally, we define global obedience slackness as $\kappa^* := \min_{\text{non-empty } \hat{\Omega} \subseteq \Omega} \kappa(\hat{\Omega})$.[34]

Proposition 2. *If the value of randomization under commitment is strictly smaller than $\kappa^* \delta$, then the value of commitment is smaller than the value of randomization under commitment.*

The bound in Proposition 2 identifies two parameters that restrict the quantitative relationship between values of randomization and commitment: global obedience slackness κ^* and the smallest payoff gap δ. If the value of randomization is smaller than the product of these two parameters, then commitment cannot be more valuable than randomization. Moreover, since generically, $\kappa^* \delta > 0$, Proposition 2 generalizes the only-if direction of Theorem 1.

Proposition 2 also clarifies the key ingredients in the aforementioned examples in Online Appendix B.3.1. A large value of commitment can coexist with an arbitrarily small value of randomization only in environments where: (i) Receiver becomes nearly indifferent at a partitional posterior (κ^* is small) or (ii) Sender becomes nearly indifferent between two actions in some state (δ is small).

[33] We assume that the denominator is strictly positive; otherwise, the values of commitment and randomization are both trivially zero. Note that $\kappa(\hat{\Omega})$ does not depend on the choice of a^* because if $|A^*\left(\hat{\Omega}\right)| \neq 1$, $\kappa(\hat{\Omega})$ is zero.

[34] Observe that when the environment satisfies partitional-unique-response, both $A^R(\hat{\Omega})$ and $A^*(\hat{\Omega})$ are singletons for any non-empty $\hat{\Omega}$, which implies $\kappa^* > 0$. Conversely, if $\kappa^* > 0$, the environment satisfies partitional-unique-response.

Away from such degenerate cases, small value of randomization implies that commitment is even less valuable than randomization.

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A Appendix

A.1 Notation and terminology

Let $A = \{a_1, ..., a_{|A|}\}$. Let $\Omega = \{\omega_1, ..., \omega_{|\Omega|}\}$.

Given a messaging strategy σ, let $M_\sigma = \{m \in M | \sigma(m|\omega) > 0 \text{ for some } \omega\}$ be the set of messages that are sent with positive probability under σ. For any ω, if $\sigma(\cdot|\omega)$ is degenerate (i.e., there exists a message m such that $\sigma(m|\omega) = 1$), let $\sigma(\omega)$ denote the message that is sent in state ω. Similarly, if $\rho(\cdot|m)$ is degenerate, let $\rho(m)$ denote the action taken following message m.

Say that ρ is *pure* if $\rho(\cdot|m)$ is degenerate for all $m \in M$. Given a profile (σ, ρ), say ρ is *pure-on-path* if $\rho(\cdot|m)$ is degenerate for all $m \in M_\sigma$.

Let $I = \begin{bmatrix} e_1 \\ ... \\ e_{|\Omega|} \end{bmatrix}$ be the identity matrix of size $|\Omega|$, with e_i being the row vector with all entries equal to 0 except for the ith entry equal to 1. Denote a vector all of whose elements are equal to r by $\boldsymbol{r}$. Denote the jth element of μ by $[\mu]_j$.

A.2 Generic environments for the proofs

We now introduce two generic sets of environments that play important roles in the proofs.

A.2.1 Partitional-unique-response environments

An environment (u_S, u_R) satisfies *partitional-unique-response* if for every non-empty $\hat{\Omega} \subseteq \Omega$,

$$\arg\max_{a \in A} \sum_{\omega \in \hat{\Omega}} \mu_0(\omega) u_R(a, \omega)$$

is a singleton.

Note that whether an environment satisfies partitional-unique-response does not depend on Sender's preferences. The partitional-unique-response property requires that, at the finitely many beliefs that can be induced by a partitional experiment, Receiver has a unique best response.

Lemma 1. *The set of partitional-unique-response environments is generic.*

Proof. Let $\mathcal{T}$ denote the set of triplets $\left(\hat{\Omega}, a_i, a_j\right)$ such that $\emptyset \neq \hat{\Omega} \subseteq \Omega$, $a_i, a_j \in A$, and $a_i \neq a_j$. For any $\left(\hat{\Omega}, a_i, a_j\right) \in \mathcal{T}$, let $Q(\hat{\Omega}, a_i, a_j)$ denote the set of u_R such that

$$\sum_{\omega \in \hat{\Omega}} \mu_0(\omega) u_R(a_i, \omega) = \sum_{\omega \in \hat{\Omega}} \mu_0(\omega) u_R(a_j, \omega). \tag{2}$$

We will show that $\cup_{\left(\hat{\Omega}, a_i, a_j\right) \in \mathcal{T}} Q(\hat{\Omega}, a_i, a_j)$ has measure zero in $[0,1]^{|A||\Omega|}$. Since A and Ω are finite, it suffices to show that for any $\left(\hat{\Omega}, a_i, a_j\right) \in \mathcal{T}$, $Q(\hat{\Omega}, a_i, a_j)$ has measure zero.

Fix any $\left(\hat{\Omega}, a_i, a_j\right) \in \mathcal{T}$. Note that $Q(\hat{\Omega}, a_i, a_j)$ can be written as

$$\left\{ u_R \in [0,1]^{|A||\Omega|} \mid \sum_{\omega, a} u_R(a, \omega) \eta(a, \omega) = 0 \right\} \tag{3}$$

where

$$\eta(a, \omega) = \begin{cases} \mu_0(\omega) & \text{if } a = a_i, \omega \in \hat{\Omega} \\ -\mu_0(\omega) & \text{if } a = a_j, \omega \in \hat{\Omega} \\ 0 & \text{otherwise.} \end{cases}$$

Hence, $Q(\hat{\Omega}, a_i, a_j)$ is a subset of a hyperplane in $\mathbb{R}^{|A||\Omega|}$, and thus has measure zero.[35] □

A.2.2 Scant-indifferences environments

For each $a_i \in A$, let $\mathbf{u}_S(a_i) = u_S(a_i, \cdot) \in \mathbb{R}^{|\Omega|}$ and $\mathbf{u}_R(a_i) = u_R(a_i, \cdot) \in \mathbb{R}^{|\Omega|}$ denote the payoff vectors across states.

For each a_i, define the *expanded-indifference matrix* T^i as follows. Let T^i_S be the matrix with $|A| - 1$ rows and $|\Omega|$ columns, with each row associated with $j \neq i$ and equal to $\mathbf{u}_S(a_j) - \mathbf{u}_S(a_i)$. Let T^i_R be the matrix with $|A| - 1$ rows and $|\Omega|$ columns, with each row associated with $j \neq i$ and

[35] The set of partitional-unique-response environments is also generic in the topological sense. It is easy to see that $Q(\hat{\Omega}, a_i, a_j)$ is closed. Since $\cup_{\left(\hat{\Omega}, a_i, a_j\right) \in \mathcal{T}} Q(\hat{\Omega}, a_i, a_j)$ is therefore closed, its complement is open. Since $\cup_{\left(\hat{\Omega}, a_i, a_j\right) \in \mathcal{T}} Q(\hat{\Omega}, a_i, a_j)$ has measure zero, its complement is dense. Thus, the set of partitional-unique-response environments, which is a superset of the complement of $\cup_{\left(\hat{\Omega}, a_i, a_j\right) \in \mathcal{T}} Q(\hat{\Omega}, a_i, a_j)$, contains an open, dense set.

equal to $\mathbf{u}_R(a_j) - \mathbf{u}_R(a_i)$. Then, let

$$T^i = \begin{bmatrix} T_S^i \\ T_R^i \\ I \end{bmatrix}.$$

Given any matrix T, a *row-submatrix* of T is a matrix formed by removing some of the rows of T.

We say that an environment satisfies *scant-indifferences* if for each $a_i \in A$, every row-submatrix of the expanded-indifference matrix T^i is full rank.

We anticipate that the reader might find this definition mysterious, so we now try to provide some intuition by connecting this definition to the proof sketch we gave in the body of the paper for Theorem 2 in the case with two actions and three states.

Recall that, in Figure 1, the argument behind Lemma 4 relied on two facts that must hold generically. First, the border between R_1 and R_2 and the border between S_1 and S_2 have at most one intersection, μ_m. Second, generically μ_m (if it exists) is an interior belief. Moreover, the argument behind Lemma 5 relied on the fact that, generically, for any ω and $a_i \neq a_j$, $u_S(a_i, \omega) \neq u_S(a_j, \omega)$.

We now illustrate why these three facts hold in any scant-indifferences environment. With only two actions, we can look at T^1 only, since the argument for T^2 is identical. We have

$$T^1 = \begin{bmatrix} \overset{\Delta}{u}_S(\omega_1) & \overset{\Delta}{u}_S(\omega_2) & \overset{\Delta}{u}_S(\omega_3) \\ \overset{\Delta}{u}_R(\omega_1) & \overset{\Delta}{u}_R(\omega_2) & \overset{\Delta}{u}_R(\omega_3) \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

where $\overset{\Delta}{u}_S(\omega_i) = u_S(a_2, \omega_i) - u_S(a_1, \omega_i)$ and analogously for $\overset{\Delta}{u}_R$.

First, consider the row-submatrix

$$\overset{\Delta}{T} = \begin{bmatrix} \overset{\Delta}{u}_S(\omega_1) & \overset{\Delta}{u}_S(\omega_2) & \overset{\Delta}{u}_S(\omega_3) \\ \overset{\Delta}{u}_R(\omega_1) & \overset{\Delta}{u}_R(\omega_2) & \overset{\Delta}{u}_R(\omega_3) \end{bmatrix}.$$

Note that both Sender and Receiver are indifferent between the two actions at a belief μ if and only if $\overset{\Delta}{T}\mu = 0$. If $\overset{\Delta}{T}$ is full-rank, then $\{x : \overset{\Delta}{T}x = 0\}$ is a line through the origin. This line intersects the belief simplex $\{\mu : 1 \cdot \mu = 1, \mu \geq 0\}$ at most once. Therefore, in a scant-indifferences environment, the border between R_1 and R_2 and the border between S_1 and S_2 have at most one intersection.

Second, consider the row-submatrix

$$\begin{bmatrix} \overset{\Delta}{u}_S(\omega_1) & \overset{\Delta}{u}_S(\omega_2) & \overset{\Delta}{u}_S(\omega_3) \\ \overset{\Delta}{u}_R(\omega_1) & \overset{\Delta}{u}_R(\omega_2) & \overset{\Delta}{u}_R(\omega_3) \\ 1 & 0 & 0 \end{bmatrix}.$$

Requiring that this matrix be full-rank yields that μ_m puts strictly positive probability on ω_1. Considering the row-submatrices that alternatively include the other two rows of the identity matrix yields that μ_m puts strictly positive probability on ω_2 and ω_3.

Finally, suppose that in, say state ω_1, $\overset{\Delta}{u}_S(\omega_1) = 0$. Consider the row-submatrix

$$\begin{bmatrix} 0 & \overset{\Delta}{u}_S(\omega_2) & \overset{\Delta}{u}_S(\omega_3) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Clearly, this matrix is not full-rank, so scant-indifferences rules out the possibility that $u_S(a_1, \omega_1) = u_S(a_2, \omega_1)$.

Having motivated the definition of scant-indifferences environments, we now establish that the set of such environments is generic.

Lemma 2. *The set of scant-indifferences environments is generic.*

Proof. First, observe that given any expanded-indifference matrix T^i, if every square row-submatrix of T^i is full-rank, then every row-submatrix of T^i is full-rank. To see why, suppose every square row-submatrix of T^i is full-rank. Now, consider an arbitrary row-submatrix $\hat{T}$ of T^i. If $\hat{T}$ is square, it obviously has full rank. Suppose that $\hat{T}$ has more than $|\Omega|$ rows. In that case, every square row-submatrix of $\hat{T}$ is also a square row-submatrix of T^i. Any such row-submatrix has rank $|\Omega|$. Therefore, $\hat{T}$ has rank $|\Omega|$ and is thus full-rank. Finally, suppose that $\hat{T}$ has fewer than $|\Omega|$ rows.

We know that $\hat{T}$ is a row-submatrix of some square row-submatrix $\tilde{T}$ of T^i. We know $\tilde{T}$ has full rank so all of its rows are linearly independent. Consequently, the subset of its rows that constitute $\hat{T}$ is also linearly independent.

Hence, we can consider only square row-submatrices of T^i. Recall that a square matrix is full-rank if and only if its determinant is non-zero. Thus, it will suffice to show that for a full Lebesgue measure set of (u_S, u_R), the determinant of every square row-submatrix of each expanded-indifference matrix is non-zero. Consider any square row-submatrix $\hat{T}$ of any expanded-indifference matrix. The determinant of $\hat{T}$ is a non-zero polynomial function of (u_S, u_R).[36] The zero set of any non-zero polynomial function has Lebesgue measure zero, so the set of (u_S, u_R) for which $\hat{T}$ does not have full rank is a measure-zero set. Since there are only finitely many square row-submatrices of expanded-indifference matrices, the set of scant-indifferences environments is generic.[37] □

As we noted above (for the three-state, two-action case), in scant-indifferences environments, there is no state in which Sender is indifferent between two distinct actions.

Lemma 3. *In any scant-indifferences environment, for any ω and $a_i \neq a_j$, $u_S(a_i, \omega) \neq u_S(a_j, \omega)$.*

Proof. Suppose, toward a contradiction, that there exist some ω, a_i, and a_j such that $u_S(a_i, \omega) = u_S(a_j, \omega)$. Without loss, suppose this holds for ω_1. Then, the vector $\mathbf{u}_S(a_i) - \mathbf{u}_S(a_j)$ has zero as its first element. Now consider the $|\Omega| \times |\Omega|$ row-submatrix of T^j

$$\begin{bmatrix} \mathbf{u}_S(a_i) - \mathbf{u}_S(a_j) \\ e_2 \\ \dots \\ e_{|\Omega|} \end{bmatrix}.$$

[36] To establish that this polynomial function is not identically zero, we need to show that there are values of (u_S, u_R) such that the determinant of $\hat{T}$ is not zero. Recall that rearranging the rows of a matrix does not change whether the determinant is zero. Let $\tilde{T}$ be some matrix constructed by rearranging the rows of $\hat{T}$ so that if $\tilde{T}$ contains any row equal to e_k, that row is the kth row of $\tilde{T}$. Any other row ℓ of $\tilde{T}$ must belong either to T^i_S or T^i_R. Construct (u_S, u_R) as follows. If $\mathbf{u}_S(a_j) - \mathbf{u}_S(a_i)$ is not in $\hat{T}$ (and thus not in $\tilde{T}$), set $\mathbf{u}_S(a_j)$ in an arbitrary way. If $\mathbf{u}_S(a_j) - \mathbf{u}_S(a_i)$ is the ℓth row of $\tilde{T}$, set $\mathbf{u}_S(a_j)$ as the row vector with all entries equal to 0 except for the ℓth entry equal to 1. Similarly, if $\mathbf{u}_R(a_j) - \mathbf{u}_R(a_i)$ is not in $\hat{T}$ (and thus not in $\tilde{T}$), set $\mathbf{u}_R(a_j)$ in an arbitrary way, and if $\mathbf{u}_R(a_j) - \mathbf{u}_R(a_i)$ is the ℓth row of $\tilde{T}$, set $\mathbf{u}_R(a_j)$ as the row vector with all entries equal to 0 except for the ℓth entry equal to 1. Finally, set $\mathbf{u}_S(a_i) = \mathbf{u}_R(a_i) = \mathbf{0}$. With (u_S, u_R) defined this way, $\tilde{T}$ is an identity matrix with determinant 1. Therefore, the determinant of $\hat{T}$ is a non-zero function of (u_S, u_R).

[37] The zero set of any non-zero polynomial function is closed, so the set of scant-indifferences environments is generic in the topological sense as well.

This matrix is not full-rank because the first row can be expressed as a linear combination of the other rows. □

A.3 Key proposition

In this section we establish a key proposition that will be useful for proofs of Theorems 1, 2, and 3.

Proposition 3. *In a scant-indifferences environment, if commitment has no value, then there is a partitional $\hat{\sigma}$ and a pure strategy $\hat{\rho}$ such that: (i) $(\hat{\sigma}, \hat{\rho})$ is a cheap-talk equilibrium and yields the persuasion payoff, and (ii) $|M_{\hat{\sigma}}| \leq |A|$.*

To establish the Proposition, we first show that if a cheap-talk equilibrium (in fact any R-BR profile) yields the persuasion payoff, then Receiver must not randomize on path in that equilibrium (Lemma 4). Second, we show that if Receiver does not randomize on path, Sender also need not randomize (Lemma 5).

Lemma 4. *In a scant-indifferences environment, if (σ, ρ) is R-BR and yields the persuasion payoff, then ρ must be pure-on-path.*

Proof. Suppose by contradiction that the environment satisfies scant-indifferences, profile (σ, ρ) is R-BR and yields the persuasion payoff, yet there exists a message $m \in M_\sigma$ such that $|\operatorname{Supp}(\rho(\cdot|m))| = k > 1$.

We first note that both Sender and Receiver must be indifferent among all the actions in $\operatorname{Supp}(\rho(\cdot|m))$ given μ_m, the belief induced by message m. In other words, for all $a_i, a_j \in \operatorname{Supp}(\rho(\cdot|m))$,

$$\sum_\omega \mu_m(\omega) u_R(a_i, \omega) = \sum_\omega \mu_m(\omega) u_R(a_j, \omega), \tag{4}$$

$$\sum_\omega \mu_m(\omega) u_S(a_i, \omega) = \sum_\omega \mu_m(\omega) u_S(a_j, \omega). \tag{5}$$

Equation (4) follows immediately from R-BR. Equation (5) follows from the fact that (σ, ρ) yields the persuasion payoff: if, say $\sum_\omega \mu_m(\omega) u_S(a_i, \omega) > \sum_\omega \mu_m(\omega) u_S(a_j, \omega)$, an alternative strategy profile where Receiver breaks ties in favor of Sender would still satisfy R-BR while strictly improving Sender's payoff.

For each belief $\mu \in \Delta\Omega$, let $A_R^*(\mu)$ denote the set of Receiver-optimal actions under belief μ; that is, $A_R^*(\mu) = \arg\max_{a \in A} \mathbf{u}_R(a) \cdot \mu$. Clearly, $\text{Supp}(\rho(\cdot|m)) \subseteq A_R^*(\mu_m)$, meaning that $A_R^*(\mu_m)$ contains the k actions in the support of $\rho(\cdot|m)$, but may also contain additional actions that are not played following m. Without loss of generality, let $\text{Supp}(\rho(\cdot|m)) = \{a_1, ..., a_k\}$ and $A_R^*(\mu_m) = \{a_1, ..., a_k, a_{k+1}, ..., a_{k+r}\}$ for some $r \geq 0$. Note that for any $i = 2, ..., k+r$, $\mathbf{u}_R(a_1) \cdot \mu_m = \mathbf{u}_R(a_i) \cdot \mu_m$.

Equation (5) implies that for any $i = 2, ..., k$, $\mathbf{u}_S(a_1) \cdot \mu_m = \mathbf{u}_S(a_i) \cdot \mu_m$. Combining both Sender's and Receiver's indifference conditions, we have

$$\begin{bmatrix} \mathbf{u}_S(a_2) - \mathbf{u}_S(a_1) \\ ... \\ \mathbf{u}_S(a_k) - \mathbf{u}_S(a_1) \\ \\ \mathbf{u}_R(a_2) - \mathbf{u}_R(a_1) \\ ... \\ \mathbf{u}_R(a_{k+r}) - \mathbf{u}_R(a_1) \end{bmatrix} \mu_m = \mathbf{0}. \tag{6}$$

Let $\hat{\Omega} = \{\omega | \mu_m(\omega) = 0\}$, the (potentially empty) set of states that are not in the support of μ_m. Without loss, suppose that $\hat{\Omega} = \{\omega_1, ..., \omega_\ell\}$ where $\ell \geq 0$. If $\ell > 0$ (i.e., $\hat{\Omega} \neq \emptyset$), then we have

$$\begin{bmatrix} e_1 \\ ... \\ e_\ell \end{bmatrix} \mu_m = \mathbf{0}. \tag{7}$$

Let $\hat{T}_S = \begin{bmatrix} \mathbf{u}_S(a_2) - \mathbf{u}_S(a_1) \\ ... \\ \mathbf{u}_S(a_k) - \mathbf{u}_S(a_1) \end{bmatrix}$, $\hat{T}_R = \begin{bmatrix} \mathbf{u}_R(a_2) - \mathbf{u}_R(a_1) \\ ... \\ \mathbf{u}_R(a_{k+r}) - \mathbf{u}_R(a_1) \end{bmatrix}$, $\hat{E} = \begin{bmatrix} e_1 \\ ... \\ e_\ell \end{bmatrix}$, and $\hat{T} = \begin{bmatrix} \hat{T}_S \\ \hat{T}_R \\ \hat{E} \end{bmatrix}$. Note that $\hat{T}$ is a row-submatrix of the expanded-indifference matrix T^1.

Combining (6) and (7), we know $\hat{T}\mu_m = \mathbf{0}$. Moreover, since $\mu_m \in \Delta\Omega$, we know $\mathbf{1}\mu_m = 1$.

Next we make two observations: (i) $\text{rank}(\hat{T}) < |\Omega|$, otherwise the unique solution to $\hat{T}\mu = \mathbf{0}$ is $\mu = \mathbf{0}$. Since we are in a scant-indifferences environment, this means that $\hat{T}$ has full row rank; (ii)

vector $\mathbf{1}$ cannot be represented as a linear combination of rows of $\hat{T}$. To see why, assume toward contradiction that there exists a row vector $z \in \mathbb{R}^{2k+r+\ell-2}$ such that $z\hat{T} = \mathbf{1}$. This would lead to the contradiction $1 = \mathbf{1}\mu_m = z\hat{T}\mu_m = z\mathbf{0} = 0$.

Observations (i) and (ii) together imply that the matrix $\begin{bmatrix} \hat{T} \\ \mathbf{1} \end{bmatrix}$ has full row rank. Consequently, we know $\operatorname{rank}\left(\begin{bmatrix} \hat{T} \\ \mathbf{1} \end{bmatrix}\right) > \operatorname{rank}\left(\begin{bmatrix} \hat{T}_R \\ \hat{E} \\ \mathbf{1} \end{bmatrix}\right)$.

Now, we claim that there exists $x \in \mathbb{R}^{|\Omega|}$ such that

$$\begin{bmatrix} \hat{T}_R \\ \hat{E} \\ \mathbf{1} \end{bmatrix} x = 0 \tag{8}$$

and

$$\hat{T}_S\, x \neq 0. \tag{9}$$

To see this, suppose by contradiction that for any x that solves (8), we have $\hat{T}_S\, x = 0$. This would imply that the set of solutions to (8) and the set of solutions to

$$\begin{bmatrix} \hat{T} \\ \mathbf{1} \end{bmatrix} x = 0 \tag{10}$$

coincide. By the Rank-Nullity Theorem, however, the subspace defined by (10) has dimension $|\Omega| - \operatorname{rank}\left(\begin{bmatrix} \hat{T} \\ \mathbf{1} \end{bmatrix}\right)$, while the subspace defined by (8) has a higher dimension $|\Omega| - \operatorname{rank}\left(\begin{bmatrix} \hat{T}_R \\ \hat{E} \\ \mathbf{1} \end{bmatrix}\right)$.

Consider an x that satisfies (8) and (9). Consider two vectors, $\mu_m + \varepsilon x$ and $\mu_m - \varepsilon x$, for some $\varepsilon \in \mathbb{R}_{>0}$. First we verify that for sufficiently small ε, $\mu_m \pm \varepsilon x \in \Delta\Omega$. Since $\mathbf{1}x = 0$, it follows that $\mathbf{1}\left(\mu_m \pm \varepsilon x\right) = \mathbf{1}\mu_m = 1$. For $\omega_j \notin \hat{\Omega}$, we have $[\mu_m]_j > 0$, so for small enough ε, $[\mu_m \pm \varepsilon x]_j \geq 0$. For $\omega_j \in \hat{\Omega}$, we know e_j is a row of $\hat{E}$, so $e_j x = 0$. Consequently, $[\mu_m \pm \varepsilon x]_j = e_j\left(\mu_m \pm \varepsilon x\right) = [\mu_m]_j = 0$. Thus, $\mu_m \pm \varepsilon x \in \Delta\Omega$.

Observe that $A_R^*(\mu_m) = A_R^*(\mu_m \pm \varepsilon x)$. First, for any $a \notin A_R^*(\mu_m)$, if ε is sufficiently small, $a \notin A_R^*(\mu_m \pm \varepsilon x)$. Therefore, $A_R^*(\mu_m \pm \varepsilon x) \subseteq A_R^*(\mu_m)$. But, $\hat{T}_R\, x = 0$ implies that $(\mu_m \pm \varepsilon x) \cdot \mathbf{u}_R(a)$ is constant across $a \in A_R^*(\mu_m)$, so $A_R^*(\mu_m \pm \varepsilon x) = A_R^*(\mu_m)$.

Consider an alternative messaging strategy $\hat{\sigma}$ that is identical to σ, except that the message m is split into two new messages, m^+ and m^-, which induce the beliefs $\mu_m + \varepsilon x$ and $\mu_m - \varepsilon x$, respectively.[38] We consider $\hat{\rho}$ that agrees with ρ on messages other than $\{m, m^+, m^-\}$ and leads Receiver to break indifferences in Sender's favor following m^+ and m^-. We will show that $(\hat{\sigma}, \hat{\rho})$ yields a strictly higher payoff to Sender, thus contradicting the assumption that (σ, ρ) yields the persuasion payoff.

Since $\hat{T}_S\, x \neq 0$, we know there is an $a_i \in \{a_2, ..., a_k\}$ such that $x \cdot (\mathbf{u}_S(a_i) - \mathbf{u}_S(a_1)) \neq 0$.

Because $a_1 \in A_R^*(\mu_m \pm \varepsilon x) = A_R^*(\mu_m)$, we have

$$\max_{a \in A_R^*(\mu_m)} (\mu_m + \varepsilon x) \cdot (\mathbf{u}_S(a) - \mathbf{u}_S(a_1)) \geq 0$$

and

$$\max_{a \in A_R^*(\mu_m)} (\mu_m - \varepsilon x) \cdot (\mathbf{u}_S(a) - \mathbf{u}_S(a_1)) \geq 0.$$

We now establish that at least one of these inequalities has to be strict. Suppose toward contradiction that both hold with equality. The first equality implies that for all $a_i \in \{a_2, ..., a_k\}$, $(\mu_m + \varepsilon x) \cdot (\mathbf{u}_S(a_i) - \mathbf{u}_S(a_1)) \leq 0$, which combined with the fact that $\mu_m \cdot \mathbf{u}_S(a_i) = \mu_m \cdot \mathbf{u}_S(a_1)$ implies that $x \cdot (\mathbf{u}_S(a_i) - \mathbf{u}_S(a_1)) \leq 0$ for all $a_i \in \{a_2, ..., a_k\}$. Similarly, the second equality implies that $-x \cdot (\mathbf{u}_S(a_i) - \mathbf{u}_S(a_1)) \leq 0$ for all $a_i \in \{a_2, ..., a_k\}$. Together, this yields that $x \cdot (\mathbf{u}_S(a_i) - \mathbf{u}_S(a_1)) = 0$ for all $a_i \in \{a_2, ..., a_k\}$, a contradiction. Hence, one of the inequalities has to be strict.

[38] It is possible for $M_\sigma = M$, but we can consider an alternative strategy that induces the same outcome as σ and uses only $|A|$ messages. We can also let m play the role of m^+ or m^-, so our assumption that $|M| \geq |A| + 1$ suffices.

Consequently, Sender's interim payoff under $\hat{\sigma}$ (in the event that m is sent under σ) is

$$\begin{aligned}
&\frac{1}{2}\max_{a\in A_R^*(\mu_m)}(\mu_m+\varepsilon x)\cdot\mathbf{u}_S(a)+\frac{1}{2}\max_{a\in A_R^*(\mu_m)}(\mu_m-\varepsilon x)\cdot\mathbf{u}_S(a)\\
>&\frac{1}{2}(\mu_m+\varepsilon x)\cdot\mathbf{u}_S(a_1)+\frac{1}{2}(\mu_m-\varepsilon x)\cdot\mathbf{u}_S(a_1)\\
=&\mu_m\cdot\mathbf{u}_S(a_1)
\end{aligned}$$

Thus, $(\hat{\sigma},\hat{\rho})$ yields a strictly higher payoff to Sender, contradicting the assumption that (σ,ρ) yields the persuasion payoff. □

Lemma 5. *In a scant-indifferences environment, if a cheap-talk equilibrium (σ,ρ) yields the persuasion payoff and ρ is pure-on-path, then there exists a partitional $\hat{\sigma}$ and a pure strategy $\hat{\rho}$ such that $|M_{\hat{\sigma}}|\leq|A|$ and $(\hat{\sigma},\hat{\rho})$ is a cheap-talk equilibrium and yields the persuasion payoff.*

Proof. Suppose a cheap-talk equilibrium (σ,ρ) yields the persuasion payoff and ρ is pure-on-path.

First, we show that for any ω and any m,m' such that $\sigma(m|\omega),\sigma(m'|\omega)>0$, $\rho(m)=\rho(m')$. The fact that both m and m' are sent in ω implies, by S-BR, that $u_S(\rho(m),\omega)=u_S(\rho(m'),\omega)$. Moreover, by Lemma 3, there exist no distinct a and a' such that $u_S(a,\omega)=u_S(a',\omega)$, so it must be that $\rho(m)=\rho(m')$.

Let $A^*=\{a\in A|a=\rho(m)\text{ for some }m\in M_\sigma\}$ be the set of actions that are taken on-path. Without loss, let $A^*=\{a_1,...,a_k\}$. For each a_i, let $M_i=\{m\in M_\sigma|\rho(m)=a_i\}$ be the set of on-path messages that induce action a_i, and $\Omega_i=\{\omega\in\Omega|\,\mathrm{Supp}(\sigma(\cdot|\omega))\subseteq M_i\}$ be the set of states that induce action a_i. Note that $\{M_i\}_{i=1}^k$ is a partition of M_σ. Moreover, it is easy to see that $\{\Omega_i\}_{i=1}^k$ is a partition of Ω. First, Ω_i cannot be empty because every $a_i\in A^*$ is taken on-path. Second, every $\omega\in\Omega$ belongs to some Ω_i as only actions in A^* are taken on-path; hence, $\cup_i\Omega_i=\Omega$. Finally, the fact that for any ω and any m,m' such that $\sigma(m|\omega),\sigma(m'|\omega)>0$ we have $\rho(m)=\rho(m')$ implies that if $i\neq j$, Ω_i and Ω_j are disjoint.

Now select one message in each M_i, and label it as m_i. Consider the following alternative strategy profile $(\hat{\sigma},\hat{\rho})$:

- $\hat{\sigma}(m_i|\omega)=1$ if $\omega\in\Omega_i$.
- $\hat{\rho}(m_i)=a_i$.

- $\hat{\rho}(m) = a_1$ if $m \in M \backslash \{m_1, ..., m_k\}$.

Note that $\hat{\sigma}$ is well-defined because $\{\Omega_i\}_{i=1}^k$ is a partition of Ω. By construction, $\hat{\sigma}$ is partitional, $|M_{\hat{\sigma}}| \leq |A|$, and $\hat{\rho}$ is a pure strategy. Moreover, under both (σ, ρ) and $(\hat{\sigma}, \hat{\rho})$, every state in Ω_i induces action a_i with probability 1. Thus, the two strategy profiles induce the same distribution over states and actions, so $(\hat{\sigma}, \hat{\rho})$ also yields the persuasion payoff. It remains to show that $(\hat{\sigma}, \hat{\rho})$ is a cheap-talk equilibrium.

Note that S-BR of (σ, ρ) implies that for any ω and $m \in \text{Supp}(\sigma(\cdot|\omega))$, we have

$$u_S(\rho(m), \omega) \geq u_S(\rho(m'), \omega) \text{ for all } m' \in M_\sigma.$$

Therefore, for any $\omega \in \Omega_i$, $u_S(a_i, \omega) \geq u_S(a_j, \omega)$ for all $a_j \in A^*$. This implies that $u_S(\hat{\rho}(\hat{\sigma}(\omega)), \omega) \geq u_S(\hat{\rho}(m'), \omega)$ for all $m' \in M$. Hence, $(\hat{\sigma}, \hat{\rho})$ satisfies S-BR.

The fact that (σ, ρ) is R-BR implies that for all $m \in M_\sigma$,

$$\sum_{\omega \in \Omega} \mu_0(\omega)\sigma(m|\omega)u_R(\rho(m), \omega) \geq \sum_{\omega \in \Omega} \mu_0(\omega)\sigma(m|\omega)u_R(a', \omega) \quad \text{for all } a' \in A.$$

For any $i \in \{1, ..., k\}$, we can sum the inequality above over $m \in M_i$. Since for $m \in M_i$ we have $\rho(m) = a_i$, this yields

$$\sum_{\omega \in \Omega} \mu_0(\omega) \sum_{m \in M_i} \sigma(m|\omega)u_R(a_i, \omega) \geq \sum_{\omega \in \Omega} \mu_0(\omega) \sum_{m \in M_i} \sigma(m|\omega)u_R(a', \omega) \quad \text{for all } a' \in A.$$

Since for any $m \in M_i$ and $\omega \notin \Omega_i$, we have $\sigma(m|\omega) = 0$, the inequality above implies

$$\sum_{\omega \in \Omega_i} \mu_0(\omega) \sum_{m \in M_i} \sigma(m|\omega)u_R(a_i, \omega) \geq \sum_{\omega \in \Omega_i} \mu_0(\omega) \sum_{m \in M_i} \sigma(m|\omega)u_R(a', \omega) \quad \text{for all } a' \in A.$$

Since $\sum_{m \in M_i} \sigma(m|\omega) = 1$ if $\omega \in \Omega_i$, we have

$$\sum_{\omega \in \Omega_i} \mu_0(\omega)u_R(a_i, \omega) \geq \sum_{\omega \in \Omega_i} \mu_0(\omega)u_R(a', \omega) \quad \text{for all } a' \in A. \tag{11}$$

To establish $(\hat{\sigma}, \hat{\rho})$ is R-BR, we need to show that for any $m_i \in M_{\hat{\sigma}}$, we have

$$\sum_{\omega \in \Omega} \mu_0(\omega)\hat{\sigma}\left(m_i|\omega\right) \sum_{a \in A} \hat{\rho}\left(a|m_i\right) u_R(a,\omega) \geq \sum_{\omega \in \Omega} \mu_0(\omega)\hat{\sigma}\left(m_i|\omega\right) u_R(a',\omega) \quad \text{for all } a' \in A.$$

But, by definition of $(\hat{\sigma}, \hat{\rho})$, we know that $\hat{\sigma}\left(m_i|\omega\right) = 0$ for $\omega \notin \Omega_i$ and that $\hat{\rho}\left(a_i|m_i\right) = 1$. Hence, the inequality above is equivalent to Equation (11). □

A.4 Proof of Theorem 1

Here we present and prove a result that generalizes Theorem 1 into a threefold equivalence.

Theorem 1′. *Generically, the following statements are equivalent:*

(i) Commitment is valuable.

(ii) Committed Sender values randomization.

(iii) For any optimal persuasion profile (σ, ρ), there exists $m \in M_\sigma$ such that

$$|\arg\max_{a \in A} \sum_{\omega} \mu_m(\omega) u_R(a,\omega)| \geq 2,$$

where μ_m is defined as $\mu_m(\omega) = \frac{\mu_0(\omega)\sigma(m|\omega)}{\sum_{\omega'} \mu_0(\omega')\sigma(m|\omega')}$.

Proof. We establish the equivalence for any environment that satisfies both partitional-unique-response and scant-indifferences. Since the set of partitional-unique-response environments is generic (Lemma 1) and the set of scant-indifferences environments is generic (Lemma 2), the set of environments that satisfy both properties is also generic.

We will establish that (ii) implies (i), then that (i) implies (iii), and finally that (iii) implies (ii).

Since we are in a scant-indifferences environment, (ii) implies (i) by Proposition 3.

Next we wish to show that (i) implies (iii). We do so by establishing the contrapositive. Suppose that there exists an optimal persuasion profile (σ, ρ) such that for every $m \in M_\sigma$, $\arg\max_{a \in A} \sum_{\omega} \mu_m(\omega) u_R(a,\omega)$ is unique. This implies that ρ must be pure-on-path. We will construct an optimal persuasion profile $(\sigma, \hat{\rho})$ that is a cheap-talk equilibrium. Consider the following $\hat{\rho}$: for all $m \in M_\sigma$, let $\hat{\rho}(m) = \rho(m)$; for $m \notin M_\sigma$, let $\hat{\rho}(m) = \rho(m_0)$ for some $m_0 \in M_\sigma$. Since $\hat{\rho}$ and

ρ coincide on path, (σ, ρ) and $(\sigma, \hat{\rho})$ yield the same payoffs to both Sender and Receiver. Therefore, $(\sigma, \hat{\rho})$ satisfies R-BR and yields the persuasion payoff. It remains to show that $(\sigma, \hat{\rho})$ is S-BR, which is equivalent to Sender's interim optimality: for each ω,

$$\sum_m \sigma(m|\omega) u_S(\hat{\rho}(m), \omega) \geq u_S(\hat{\rho}(m'), \omega) \tag{12}$$

for all $m' \in M$. First, note that it suffices to show that Equation (12) holds for $m' \in M_\sigma$. Once we establish that, we know $\sum_m \sigma(m|\omega) u_S(\hat{\rho}(m), \omega) \geq u_S(\hat{\rho}(m_0), \omega)$ since $m_0 \in M_\sigma$. Therefore, since $\hat{\rho}(m') = \rho(m_0) = \hat{\rho}(m_0)$ for $m' \notin M_\sigma$, Equation (12) holds for $m' \notin M_\sigma$.

Now, suppose toward contradiction that there exist $\hat{\omega}$ and $\hat{m} \in M_\sigma$ such that $\sum_m \sigma(m|\hat{\omega}) u_S(\hat{\rho}(m), \hat{\omega}) < u_S(\hat{\rho}(\hat{m}), \hat{\omega})$. Consider an alternative messaging strategy $\hat{\sigma}$: $\hat{\sigma}(\omega) = \sigma(\omega)$ for $\omega \neq \hat{\omega}$ while $\hat{\sigma}(\hat{\omega})$ sends the same distribution of messages as $\sigma(\hat{\omega})$ with probability $1-\varepsilon$ and otherwise sends message $\hat{m}$. Formally, $\hat{\sigma}(m|\hat{\omega}) = \begin{cases} (1-\varepsilon)\,\sigma(m|\hat{\omega}) & \text{if } m \neq \hat{m} \\ (1-\varepsilon)\,\sigma(m|\hat{\omega}) + \varepsilon & \text{if } m = \hat{m} \end{cases}$.

Fix any $m \in M_\sigma$. Since A is finite, the fact that $\hat{\rho}(m) = \rho(m)$ is the unique $\arg\max_{a \in A} \sum_\omega \mu_m(\omega) u_R(a, \omega)$ implies that $\hat{\rho}(m)$ remains the best response for a neighborhood of beliefs around μ_m. Therefore, for sufficiently small ε, $(\hat{\sigma}, \hat{\rho})$ is R-BR. Hence, $(\hat{\sigma}, \hat{\rho})$ is a persuasion profile and yields the payoff

$$\begin{aligned} U_S(\hat{\sigma}, \hat{\rho}) &= U_S(\sigma, \hat{\rho}) + \varepsilon \mu_0(\hat{\omega})[u_S(\hat{\rho}(\hat{m}), \hat{\omega}) - \sum_m \sigma(m|\hat{\omega}) u_S(\hat{\rho}(m), \hat{\omega})] \\ &> U_S(\sigma, \hat{\rho}). \end{aligned}$$

This contradicts the fact that $(\sigma, \hat{\rho})$ yields the persuasion payoff.

Finally, since we are considering a partitional-unique-response environment, the fact that (iii) implies (ii) is immediate. □

A.5 Proof of Theorem 2

Lemma 2 and Proposition 3 jointly imply Theorem 2.

A.6 Proof of Theorem 3

Let λ_n denote the Lebesgue measure on $\mathbb{R}^n$. Recall that the set of environments is $[0,1]^{2|A||\Omega|}$. For any property p of an environment, let $\lambda_{2|A||\Omega|}(p) := \lambda_{2|A||\Omega|}(\{(u_S,u_R)|(u_S,u_R) \text{ satisfies } p\})$ denote the share of environments that satisfy p.

Given u_S, let $\Omega_i^{u_S} = \{\omega \in \Omega | a_i \in \arg\max_{a\in A} u_S(a,\omega)\}$ denote the set of states where a_i is an ideal action for Sender.[39] Note that each ω must belong to at least one $\Omega_i^{u_S}$, but the same ω may appear in multiple $\Omega_i^{u_S}$. Say that u_S is *regular* if $\Omega_i^{u_S} \cap \Omega_j^{u_S} = \emptyset$ for $i \neq j$. We also say that an environment (u_S,u_R) is regular if u_S is regular. Recall that an environment is felicitous if Sender obtains his ideal payoff under cheap talk. We say that an environment is regular-felicitous if it is regular and felicitous. Lemmas 2 and 3 jointly imply that $\lambda_{|A||\Omega|}(\{u_S \in [0,1]^{A\times\Omega} | u_S \text{ is regular}\}) = 1$, so $\lambda_{2|A||\Omega|}$ (regular-felicity) $= \lambda_{2|A||\Omega|}$ (felicity).

The next lemma provides a convenient characterization of regular-felicity.

Lemma 6. *A regular environment is felicitous if and only if for each i,*

$$\sum_{\omega\in\Omega_i^{u_S}} \mu_0(\omega)\left(u_R(a_i,\omega) - u_R\left(a',\omega\right)\right) \geq 0 \quad \textit{for all } a' \in A. \tag{13}$$

Proof. Suppose that u_S is regular and that (13) holds. Select $|A|$ elements from M and denote them by m_1 through $m_{|A|}$. Consider the following strategy profile (σ,ρ):

- $\sigma(\omega) = m_i$ for every $\omega \in \Omega_i^{u_S}$;
- $\rho(m) = a_i$ for $m = m_i$;
- $\rho(m) = a_1$ for $m \notin \{m_1,...,m_{|A|}\}$.

From regularity and (13), (σ,ρ) is R-BR. In addition, in every state, Sender induces his ideal action, so (σ,ρ) is S-BR and yields Sender's ideal payoff. Hence, the environment is felicitous.

Conversely, suppose that u_S is regular and that Sender obtains his ideal payoff in a cheap-talk equilibrium (σ,ρ). By regularity, Sender's ideal action in each state is unique. This implies

[39] In the body of the paper we denoted this set as Ω_i, but for the formal proofs, it is helpful to keep track of the fact that this set depends on u_S.

that ρ must be pure-on-path; otherwise, Sender would fail to induce his unique ideal action with probability 1 in some state.

For each i, let $M_i = \{m \in M_\sigma | \rho(m) = a_i\}$ be the set of on-path messages that induce action a_i. For any $m \in M_i$, R-BR implies that

$$\sum_{\omega \in \Omega} \mu_0(\omega)\sigma(m|\omega)[u_R(a_i, \omega) - u_R(a', \omega)] \geq 0 \quad \text{for all } a' \in A.$$

Summing the inequality above over $m \in M_i$, we obtain

$$\sum_{\omega \in \Omega} \sum_{m \in M_i} \mu_0(\omega)\sigma(m|\omega)[u_R(a_i, \omega) - u_R(a', \omega)] \geq 0 \quad \text{for all } a' \in A.$$

Since (σ, ρ) yields Sender's ideal payoff, we have that (i) for $\omega \in \Omega_i^{u_S}$,$\sum_{m \in M_i} \sigma(m|\omega) = 1$; and (ii) for $\omega \notin \Omega_i^{u_S}$, $\sigma(m|\omega) = 0$ for all $m \in M_i$. Therefore, the inequality simplifies to

$$\sum_{\omega \in \Omega_i^{u_S}} \mu_0(\omega)[u_R(a_i, \omega) - u_R(a', \omega)] \geq 0 \quad \text{for all } a' \in A.$$

□

A.6.1 Arbitrary state space

In this section, we establish that for any Ω, the share of environments such that commitment has no value is weakly greater than $\frac{1}{|A|^{|A|}}$.

Lemma 7. $\lambda_{2|A||\Omega|}$ (*regular-felicity*) $\geq \frac{1}{|A|^{|A|}}$.

Proof. Fix some regular u_S. For any non-empty $\Omega_i^{u_S}$, let

$$E_i = \{u_R \in [0,1]^{A \times \Omega_i^{u_S}} | \sum_{\omega \in \Omega_i^{u_S}} \mu_0(\omega) u_R(a_i, \omega) > \sum_{\omega \in \Omega_i^{u_S}} \mu_0(\omega) u_R(a', \omega), \forall a' \neq a_i\}$$

denote the set of Receiver's preferences on $A \times \Omega_i^{u_S}$ such that a_i is Receiver's uniquely optimal action given the information that $\omega \in \Omega_i^{u_S}$. By symmetry, we have that $\lambda_{|A||\Omega_i^{u_S}|}(E_i) = \frac{1}{|A|}$.[40] The

[40]Note that the set of Receiver's preferences on $A \times \Omega_i^{u_S}$ such that Receiver is indifferent between two actions, given the information that $\omega \in \Omega_i^{u_S}$, has measure zero.

set of $u_R \in [0,1]^{A\times\Omega}$ such that (u_S, u_R) is felicitous coincides, up to a set of measure zero, with $\prod_{i:\Omega_i^{u_S} \text{ is non-empty}} E_i$, whose measure is

$$\begin{aligned}
\lambda_{|A||\Omega|}\left(\prod_{i:\Omega_i^{u_S} \text{ is non-empty}} E_i\right) &= \prod_{i:\Omega_i^{u_S} \text{ is non-empty}} \lambda_{|A||\Omega_i^{u_S}|}(E_i) \\
&= \prod_{i:\Omega_i^{u_S} \text{ is non-empty}} (1/|A|) \\
&\geq \prod_{i\in\{1,\ldots,|A|\}} (1/|A|) \\
&= \frac{1}{|A|^{|A|}}. \qquad (14)
\end{aligned}$$

So, we have established that for any regular u_S, $\lambda_{|A||\Omega|}(\{u_R|(u_S,u_R) \text{ is felicitous}\}) \geq \frac{1}{|A|^{|A|}}$. Recall that $\lambda_{|A||\Omega|}(\{u_S|u_S \text{ is regular}\}) = 1$. Therefore,

$$\begin{aligned}
\lambda_{2|A||\Omega|}\,(\text{regular-felicity}) &= \int_{\{u_S|u_S \text{ is regular}\}} \int_{\{u_R\in[0,1]^{A\times\Omega}|(u_S,u_R) \text{ is felicitous}\}} d\lambda_{|A||\Omega|}\, d\lambda_{|A||\Omega|} \\
&\geq \int_{\{u_S|u_S \text{ is regular}\}} \frac{1}{|A|^{|A|}} d\lambda_{|A||\Omega|} \\
&= \frac{1}{|A|^{|A|}}.
\end{aligned}$$

□

Since commitment has no value in any felicitous environment, Lemma 7 implies that

$$\lambda_{2|A||\Omega|}(\text{commitment has no value}) \geq \frac{1}{|A|^{|A|}}.$$

A.6.2 Limit as $|\Omega| \to \infty$

In this section, we establish that as $|\Omega| \to \infty$: $(*)$ the share of felicitous environments converges to $\frac{1}{|A|^{|A|}}$, and $(**)$ generically, commitment has no value if and only if the environment is felicitous. Together, these results imply part (ii) of Theorem 3.

We first give an outline of the proof. Consider part $(**)$. We show that generically, if the

environment is jointly-inclusive,[41] then commitment having no value implies felicity (Lemma 8). We then show that as $|\Omega| \to \infty$, the share of jointly-inclusive preferences converges to one (Lemma 9). Combining these two results, we conclude that as $|\Omega| \to \infty$, generically, commitment has no value if and only if the environment is felicitous (Lemma 10). To establish part $(*)$, recall that $\lambda_{2|A||\Omega|}(\text{felicity}) \geq \frac{1}{|A|^{|A|}}$ and that the reason this is an inequality is the possibility that some $\Omega_i^{u_S}$ might be empty. We show that as $|\Omega| \to \infty$, the share of preferences such that some $\Omega_i^{u_S}$ is empty converges to zero, which implies that the share of felicitous preferences converges to $\frac{1}{|A|^{|A|}}$ (Lemma 11).

Lemma 8. *If commitment has no value in a jointly-inclusive environment that satisfies partitional-unique-response and scant-indifferences, then the environment is felicitous.*

Proof. Consider a jointly-inclusive environment that satisfies partitional-unique-response and scant-indifferences and suppose that commitment has no value. By Proposition 3, there is a partitional σ and a pure strategy ρ such that $|M_\sigma| \leq |A|$ and (σ, ρ) is a cheap-talk equilibrium and yields the persuasion payoff.

First, note that every action is induced under (σ, ρ); that is, for any $a \in A$, there exists ω such that $a = \rho(\sigma(\omega))$. To see why, suppose toward contradiction that there is an $a^* \in A$ that is not induced. Since the environment is jointly-inclusive and satisfies partitional-unique-response and scant-indifferences, there exists ω^* such that[42]

$$u_S(a^*, \omega^*) > u_S(a, \omega^*) \text{ and } u_R(a^*, \omega^*) > u_R(a, \omega^*) \text{ for all } a \neq a^*. \tag{15}$$

Since $|M_\sigma| \leq |A| < |M|$, there is an unsent message, say m^*.

Consider the strategy profile $(\hat{\sigma}, \hat{\rho})$:

- $\hat{\sigma}(\omega) = \sigma(\omega)$ for $\omega \neq \omega^*$, and $\hat{\sigma}(m|\omega^*) = \begin{cases} (1-\varepsilon) & \text{if } m = \sigma(\omega^*) \\ \varepsilon & \text{if } m = m^* \\ 0 & \text{otherwise} \end{cases}$.

[41] Recall that an environment is jointly-inclusive if for every action a, there is some state ω such that a is an ideal action for both Sender and Receiver in ω.

[42] Joint-inclusivity yields the weak inequalities. Partitional-unique-response makes the inequality strict for Receiver. Scant-indifferences, via Lemma 3, makes the inequality strict for Sender.

- $\hat{\rho}(m) = \rho(m)$ for $m \neq m^*$, and $\hat{\rho}(m^*) = a^*$.

Note that $(\hat{\sigma}, \hat{\rho})$ is R-BR for sufficiently small ε. For any $m \notin \{\sigma(\omega^*), m^*\}$, Receiver's belief upon observing m is unchanged, so $\hat{\rho}(m) = \rho(m)$ remains a best response. For $m = m^*$, (15) implies that $\hat{\rho}(m^*) = a^*$ is the best response. For $m = \sigma(\omega^*)$, the fact that the environment satisfies partitional-unique-response implies that $\hat{\rho}(m) = \rho(m)$ is the unique best response to μ_m. Moreover, since A is finite, this further implies that $\hat{\rho}(m)$ remains the best response for a neighborhood of beliefs around μ_m. Therefore, for sufficiently small ε, $\hat{\rho}(m)$ remains a best response.

Now, note that $\rho(\sigma(\omega^*)) \neq a^*$ because a^* is not induced under (σ, ρ). By (15),

$$\begin{aligned} U_S(\hat{\sigma}, \hat{\rho}) &= U_S(\sigma, \rho) + \varepsilon \mu_0(\omega^*) \left(u_S(a^*, \omega^*) - u_S(\rho(\sigma(\omega^*)), \omega^*)\right) \\ &> U_S(\sigma, \rho). \end{aligned}$$

This contradicts the fact that (σ, ρ) yields the persuasion payoff. Hence, we have established that every action is induced under (σ, ρ).

Next, we show that this fact, coupled with the maintained assumptions, implies that the environment is felicitous. Recall that (σ, ρ) is a cheap-talk equilibrium; hence for each ω,

$$u_S(\rho(\sigma(\omega)), \omega) \geq u_S(\rho(m), \omega) \text{ for all } m \in M.$$

Since every action is induced under (σ, ρ), the inequality above is equivalent to

$$u_S(\rho(\sigma(\omega)), \omega) \geq u_S(a, \omega) \text{ for all } a \in A.$$

Moreover, since the environment satisfies scant-indifferences, Lemma 3 implies that the environment is regular and

$$u_S(\rho(\sigma(\omega)), \omega) > u_S(a, \omega) \text{ for all } a \neq \rho(\sigma(\omega)). \tag{16}$$

Hence, $\Omega_i^{u_S} = \{\omega \in \Omega | \rho(\sigma(\omega)) = a_i\}$ and $\Omega_i^{u_S} \cap \Omega_j^{u_S} = \emptyset$ for $i \neq j$. Let $M_i = \{m \in M_\sigma | \rho(m) = a_i\}$.

For each i and each $m \in M_i$, R-BR of (σ, ρ) implies

$$\sum_{\omega \in \{\omega : \sigma(\omega) = m\}} \mu_0(\omega) u_R(a_i, \omega) \geq \sum_{\omega \in \{\omega : \sigma(\omega) = m\}} \mu_0(\omega) u_R(a', \omega) \text{ for all } a' \in A.$$

Summing over all $m \in M_i$, and noting that $\cup_{m \in M_i} \{\omega : \sigma(\omega) = m\} = \{\omega \in \Omega | \rho(\sigma(\omega)) = a_i\} = \Omega_i^{u_S}$, we have

$$\sum_{\omega \in \Omega_i^{u_S}} \mu_0(\omega) u_R(a_i, \omega) \geq \sum_{\omega \in \Omega_i^{u_S}} \mu_0(\omega) u_R(a', \omega) \text{ for all } a' \in A.$$

Hence, by Lemma 6, the environment is felicitous. □

Lemma 9. *Fix A. As $|\Omega| \to \infty$, $\lambda_{2|A||\Omega|}$ (joint-inclusivity) $\to 1$.*

Proof. For each $\omega \in \Omega$ and $a \in A$, let $E_{a,\omega}^S$ denote the subset of Sender's preferences on $A \times \{\omega\}$ such that action a is Sender's ideal action in state ω, i.e., $E_{a,\omega}^S = \{u_S \in [0,1]^A \, | u_S(a, \omega) \geq u_S(a', \omega), \forall a' \neq a\}$. Because the set of Sender's preferences on $A \times \{\omega\}$ such that Sender is indifferent between two actions at ω has measure zero, by symmetry we have that $\lambda_{|A|}\left(E_{a,\omega}^S\right) = 1/|A|$ for each (a, ω). Similarly, $\lambda_{|A|}\left(E_{a,\omega}^R\right) = 1/|A|$ for each (a, ω) where $E_{a,\omega}^R = \{u_R \in [0,1]^A | u_R(a, \omega) \geq u_R(a', \omega), \forall a' \neq a\}$. Now, let $E_{a,\omega} = E_{a,\omega}^S \times E_{a,\omega}^R$ denote the subset of preferences on $A \times \{\omega\}$ such that a is an ideal action for both Sender and Receiver in state ω. By the product property of the Lebesgue measure, $\lambda_{2|A|}(E_{a,\omega}) = 1/|A|^2$. Let $E_{a,\omega}^c$ denote the complement of $E_{a,\omega}$.

Let $E_a^c = \prod_\omega E_{a,\omega}^c$ denote the subset of environments in which action a is not an ideal action for both Sender and Receiver in any state. Let E_a denote the complement of E_a^c. Note that E_a is the set of environments in which action a is an ideal action for both Sender and Receiver in at least one state.

Observe that

$$\begin{aligned}\lambda_{2|A||\Omega|}(E_a) &= 1 - \lambda_{2|A||\Omega|}\left(\prod_{\omega} E^c_{a,\omega}\right)\\ &= 1 - \prod_{\omega} \lambda_{2|A|}\left(E^c_{a,\omega}\right)\\ &= 1 - \prod_{\omega}\left(1 - \frac{1}{|A|^2}\right)\\ &= 1 - \left(1 - \frac{1}{|A|^2}\right)^{|\Omega|}.\end{aligned}$$

The intersection $\cap_{a\in A} E_a$ is the set of jointly-inclusive environments: for every action a, there exists at least one state in which a is an ideal action for both Sender and Receiver. Therefore,

$$\begin{aligned}\lambda_{2|A||\Omega|}(\text{joint-inclusivity}) =& \lambda_{2|A||\Omega|}(\cap_{a\in A} E_a)\\ =& 1 - \lambda_{2|A||\Omega|}(\cup_{a\in A} E^c_a)\\ \geq& 1 - \sum_{a\in A} \lambda_{2|A||\Omega|}(E^c_a)\\ =& 1 - \sum_{a\in A}\left(1 - \lambda_{2|A||\Omega|}(E_a)\right)\\ =& 1 - \sum_{a\in A}\left(1 - \frac{1}{|A|^2}\right)^{|\Omega|}\\ =& 1 - |A|\left(1 - \frac{1}{|A|^2}\right)^{|\Omega|}\\ \to& 1 \quad \text{as } |\Omega| \to \infty.\end{aligned}$$

□

Lemma 10. *Fix A. As $|\Omega| \to \infty$, generically, commitment has no value if and only if the environment is felicitous.*

Proof. For any Ω, commitment has no value if the environment is felicitous. Hence, we only need to show that as $|\Omega| \to \infty$, generically, commitment has no value only if the environment is felicitous. Let CNV denote the set of environments in which commitment has no value, and F denote the set of felicitous environments. We need to show that as $|\Omega| \to \infty$, $\lambda_{2|A||\Omega|}(\text{CNV} \cap \text{F}^c) \to 0$.

Let *JPS* denote the set of environments that are jointly-inclusive and satisfy partitional-unique-response and scant-indifferences. We know from Lemma 8 that JPS$\cap$CNV$\cap$F$^c = \emptyset$, so $\lambda_{2|A||\Omega|}$(JPS$\cap$ CNV $\cap$ F^c) $= 0$. As $|\Omega| \to \infty$, $\lambda_{2|A||\Omega|}$ (JPS) $\to 1$ (Lemmas 1, 2, and 9), which in turn implies $\lambda_{2|A||\Omega|}$(JPS$^c \cap$ CNV $\cap$ F^c) $\to 0$. Therefore, as $|\Omega| \to \infty$, $\lambda_{2|A||\Omega|}$(CNV $\cap$ F^c) $= \lambda_{2|A||\Omega|}$(JPS $\cap$ CNV $\cap$ F^c) $+ \lambda_{2|A||\Omega|}$(JPS$^c \cap$ CNV $\cap$ F^c) $\to 0$. □

Lemma 11. *Fix A. As* $|\Omega| \to \infty$, $\lambda_{2|A||\Omega|}$(*felicity*) $\to \frac{1}{|A|^{|A|}}$.

Proof. Since regular environments are generic, it will suffice to show that $\lambda_{2|A||\Omega|}$(regular-felicity) $\to$ $\frac{1}{|A|^{|A|}}$ as $|\Omega| \to \infty$. Let $E^S = \{u_S \in [0,1]^{A\times\Omega} | \Omega_i^{u_S}$ is non-empty for all $i\}$ and $E = \{(u_S, u_R) \in [0,1]^{A\times\Omega} \times [0,1]^{A\times\Omega} | u_S \in E^S\}$.

As noted earlier in Equation (14), for any regular $u_S \in E^S$, $\lambda_{|A||\Omega|}(\{u_R | (u_S, u_R)$ is felicitous$\}) = \prod_{i:\Omega_i^{u_S} \text{ is non-empty}} \frac{1}{|A|} = \frac{1}{|A|^{|A|}}$. Therefore,

$$\begin{aligned}
\lambda_{2|A||\Omega|}(\text{regular-felicity} \cap E) &= \lambda_{2|A||\Omega|}(\{(u_S, u_R) \in [0,1]^{2|A||\Omega|} | u_S \in E^S, (u_S, u_R) \text{ is regular-felicitous}\}) \\
&= \int_{E^S} \int_{\{u_R | (u_S, u_R) \text{ is regular-felicitous}\}} d\lambda_{|A||\Omega|}\, d\lambda_{|A||\Omega|} \\
&= \int_{E^S} \frac{1}{|A|^{|A|}}\, d\lambda_{|A||\Omega|} \\
&= \frac{1}{|A|^{|A|}} \lambda_{|A||\Omega|}(E^S) \\
&= \frac{1}{|A|^{|A|}} \lambda_{2|A||\Omega|}(E).
\end{aligned}$$

Note that any jointly-inclusive environment must be contained in E. Hence, by Lemma 9, $\lambda_{2|A||\Omega|}(E) \to 1$ as $|\Omega| \to \infty$. Therefore, $\lambda_{2|A||\Omega|}$(regular-felicity) $\to \frac{1}{|A|^{|A|}}$ as $|\Omega| \to \infty$. □

Lemmas 10 and 11 jointly yield the fact that, as $|\Omega| \to \infty$, $\lambda_{2|A||\Omega|}$(commitment has no value) $\to$ $\frac{1}{|A|^{|A|}}$.

A.7 Proof of Proposition 1

A.7.1 Two Examples

We first present two examples that will provide intuition on the proof of parts (iii), (iv), (vi), and (vii).[43] In the first example, $V^{BP} = V^{MD} > V^{CT} = V^{babbling}$, where $V^{babbling}$ denotes the babbling payoff. In the second example, $V^{MD} = V^{CT} > V^{\text{partitional CT}}$, where $V^{\text{partitional CT}}$ denotes the partitional cheap-talk payoff. Neither example is knife-edge: in Online Appendix B.4, we show that these properties continue to hold for an open neighborhood of the examples, and hence for a positive share of environments.

Example 1. State $\omega \in \{0, 1\}$ with equal prior. Action $a \in \{a_1, a_2, a_3\}$. Receiver's payoff is given by

	a_1	a_2	a_3
$\omega = 0$	1	0	-2
$\omega = 1$	-2	0	1

Receiver's best response is a_1 for $\mu \in [0, \frac{1}{3}]$, a_2 for $\mu \in [\frac{1}{3}, \frac{2}{3}]$, a_3 for $\mu \in [\frac{2}{3}, 1]$.

Sender's payoff is given by

	a_1	a_2	a_3
$\omega = 0$	$-2/3$	1	$-1/2$
$\omega = 1$	$-2/3$	0	$1/2$

Sender's indirect value function is

$$V(\mu) = \begin{cases} -2/3, & \mu \in [0, \frac{1}{3}), \\ 1 - \mu, & \mu \in [\frac{1}{3}, \frac{2}{3}], \\ \mu - 1/2, & \mu \in (\frac{2}{3}, 1]. \end{cases}$$

Persuasion The persuasion problem can be solved by concavifying $V(\mu)$, which is visualized in Figure 2. The optimal persuasion profile induces posteriors 1/3 and 1 with probabilities 3/4 and

[43] For ease of numerical calculation, these two examples use payoffs outside of the unit interval. These payoffs can of course be shrunk to a subset of the unit interval through an affine transformation, without changing the relevant comparisons.

1/4, and the persuasion payoff is 5/8.

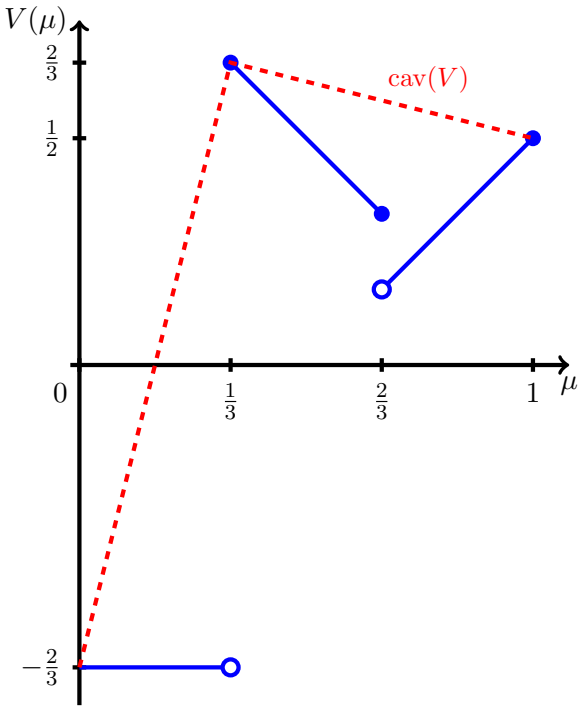


Figure 2: Concavification of $V(\mu)$

Mediation We will establish that the persuasion payoff can be achieved under mediation. By the revelation principle (see Footnote 31), it suffices to construct a state-action distribution that yields the persuasion payoff, and satisfies the truthful reporting and obedience constraints.

Consider the following $\pi : \Omega \to \Delta A$:

- $\pi(a_2|\omega = 1) = \frac{1}{2}$,
- $\pi(a_3|\omega = 1) = \frac{1}{2}$,
- $\pi(a_2|\omega = 0) = 1$.

It is easy to see that π is obedient because the posterior following action recommendation a_2 is 1/3 and the posterior following action recommendation a_3 is 1.

Now we verify π satisfies the truth-telling constraint. Note that Sender prefers action a_2 in state $\omega = 0$ and prefers action a_3 when $\omega = 1$. In state $\omega = 1$, truthful reporting leads to a mixture of a_2 and a_3, while misreporting leads to a_2. In state $\omega = 0$, truthful reporting leads to action a_2, while misreporting leads to a mixture of a_2 and a_3.

Cheap Talk Note that the babbling equilibrium yields a payoff of 1/2 to Sender. We will show that no cheap-talk equilibrium can yield a higher payoff.

Suppose toward a contradiction that there exists a cheap-talk equilibrium in which Sender's payoff is strictly greater than 1/2. Let U_ω denote Sender's expected payoff in state ω. Sender's ex-ante payoff is thus $\frac{1}{2}U_1 + \frac{1}{2}U_0$. Since $U_0 \leq 1$, for the ex-ante payoff to be strictly greater than 1/2, it must be that $U_1 > 0$. Since in equilibrium Sender is indifferent between all messages he sends in state $\omega = 1$, all such messages must yield a strictly positive payoff and thus induce action a_3 with a strictly positive probability. For Receiver to take action a_3, the posterior induced by these messages must be at least 2/3. Bayes plausibility implies that there exists some other message m inducing a posterior less than 1/2, leading to action a_1 or a_2 (or potentially a mixture of them). Therefore, message m cannot be sent in state $\omega = 1$ because it would yield a strictly lower payoff than U_1. Hence, message m is only sent in state $\omega = 0$ and induces a posterior 0 and action a_1. But then in state $\omega = 0$, Sender would like to deviate to send the messages sent in state $\omega = 1$, because any convex combination of actions a_2 and a_3 yields a strictly higher payoff than action a_1 in state $\omega = 0$. Thus, a contradiction.

Therefore, in Example 1,

$$V^{BP} = V^{MD} > V^{CT} = V^{babbling}.$$

In Online Appendix B.4.1, we show that the above value comparison holds for an open neighborhood of this example.

Example 2. State $\omega \in \{0, 1\}$ with equal prior. Action $a \in \{a_1, a_2, a_3\}$.

Receiver's payoff is given by

	a_1	a_2	a_3
$\omega = 0$	1	0	-2
$\omega = 1$	-2	0	1

Receiver's best response is a_1 for $\mu \in [0, \frac{1}{3}]$, a_2 for $\mu \in [\frac{1}{3}, \frac{2}{3}]$, a_3 for $\mu \in [\frac{2}{3}, 1]$.

Sender's payoff is given by

	a_1	a_2	a_3
$\omega = 0$	0	-1	1
$\omega = 1$	0	1	2

Partitional Cheap Talk With a binary state, a partitional cheap-talk profile is either full revelation or babbling. Full revelation cannot be an equilibrium, because Sender strictly prefers to induce action a_3 over action a_1 in both states. Babbling yields a payoff of 0 to Sender. Therefore,

$$V^{\text{partitional } CT} = V^{\text{babbling}} = 0.$$

Cheap Talk Consider the following strategy profile with two on-path messages m_1 and m_2: Sender chooses $\sigma(m_1|\omega = 0) = \sigma(m_2|\omega = 0) = 1/2$ and $\sigma(m_2|\omega = 1) = 1$; Receiver chooses $\rho(a_1|m_1) = 1$ and $\rho(a_2|m_2) = \rho(a_3|m_2) = 1/2$.

The profile satisfies R-BR because when seeing m_1 Receiver's posterior is 0 and when seeing m_2 Receiver's posterior is 2/3. The profile also satisfies S-BR: type $\omega = 0$ Sender is indifferent between sending m_1 that induces a_1 and sending m_2 that induces $\frac{1}{2}\delta_{a_2} + \frac{1}{2}\delta_{a_3}$; type $\omega = 1$ Sender prefers inducing a_2, a_3 over inducing a_1. Therefore, this profile is a cheap-talk equilibrium. Sender's payoff in this equilibrium is 0.75, which will be shown to be the mediation payoff (and thus also the cheap-talk payoff).

Mediation We show that no mediation plan can yield a higher Sender's payoff than 0.75, the payoff in the cheap-talk equilibrium constructed above.

Suppose $\pi : \Omega \to \Delta A$ is induced by some mediated equilibrium. By the revelation principle (see Footnote 31), π must satisfy obedience and truth-telling. The obedience constraints of actions a_2 and a_3 give

$$\pi(a_2|\omega = 1) \leq 2\pi(a_2|\omega = 0),$$

$$\pi(a_3|\omega = 1) \geq 2\pi(a_3|\omega = 0).$$

The two obedience constraints together imply

$$\pi(a_3|\omega=0)-\pi(a_2|\omega=0)\leq\frac{1}{2}[\pi(a_3|\omega=1)-\pi(a_2|\omega=1)].$$

Meanwhile, type 0's truth-telling constraint says

$$\pi(a_3|\omega=0)-\pi(a_2|\omega=0)\geq\pi(a_3|\omega=1)-\pi(a_2|\omega=1).$$

Combining the two inequalities above yields $\frac{1}{2}[\pi(a_3|\omega=1)-\pi(a_2|\omega=1)]\geq\pi(a_3|\omega=1)-\pi(a_2|\omega=1)$. So $\pi(a_3|\omega=1)-\pi(a_2|\omega=1)\leq 0$, and also $\pi(a_3|\omega=0)-\pi(a_2|\omega=0)\leq 0$.

Hence, Sender's expected payoff under π is

$$\frac{1}{2}[-\pi(a_2|\omega=0)+\pi(a_3|\omega=0)+\pi(a_2|\omega=1)+2\pi(a_3|\omega=1)]\leq\frac{1}{2}[\pi(a_2|\omega=1)+2\pi(a_3|\omega=1)].$$

Since $\pi(a_2|\omega=1)+\pi(a_3|\omega=1)\leq 1$ and $\pi(a_3|\omega=1)\leq\frac{1}{2}$, Sender's expected payoff under π is no more than $\frac{1}{2}[1+\frac{1}{2}]=0.75$.

Therefore, in Example 2,

$$V^{MD}=V^{CT}>V^{\text{partitional CT}}.$$

In Online Appendix B.4.2, we show that the above value comparison holds for an open neighborhood of this example.

A.7.2 Proof of the proposition

Proof. Part (i) follows immediately from the observation that the deterministic mediation payoff equals the cheap-talk payoff. Let $V^{MD,\text{deterministic}}$ denote the deterministic mediation payoff. Any cheap-talk equilibrium can be implemented by an identity communication device, so $V^{MD,\text{deterministic}}\geq V^{CT}$. Conversely, suppose (σ^*,ρ^*) is a mediated equilibrium given a deterministic communication device d. Define $\hat{\sigma}$ as $\hat{\sigma}(m_R|\omega)=\sum_{m_S\in M}d(m_R|m_S)\sigma^*(m_S|\omega)$. Choose some $m_0\in\text{Image}(d)$ and define $\hat{\rho}$ as $\hat{\rho}(\cdot|m)=\rho^*(\cdot|m)$ for $m\in\text{Image}(d)$ and $\hat{\rho}(\cdot|m)=\rho^*(\cdot|m_0)$ for $m\notin\text{Image}(d)$. It is straightforward to verify that $(\hat{\sigma},\hat{\rho})$ is a cheap-talk equilibrium, which implies $V^{MD,\text{deterministic}}\leq V^{CT}$. Hence, $V^{MD,\text{deterministic}}=V^{CT}$. Part (ii) follows from Theorem 1. Parts

(iii) and (iv) follow from Example 1 and the corresponding positive measure of environments in Online Appendix B.4.1. Part (v) follows from the neighborhood of the prosecutor-judge example. For such environments, randomization is not valuable under mediation, but $EV > 0$. To see this, note that for any small perturbation of payoffs in the prosecutor-judge example, conviction is dominant for Sender. If a state-action distribution π is induced by some mediated equilibrium, the truth-telling constraint requires that every report Sender submits must yield the same probability of conviction. Obedience then implies Receiver always chooses not to convict, yielding the babbling outcome. Therefore, $MV = 0$ and randomization is not valuable under mediation. However, commitment is obviously valuable, so $EV = CV > 0$. Part (vi) follows from Example 2 and the corresponding positive measure of environments in Online Appendix B.4.2. Part (vii) follows from Example 1 and the corresponding positive measure of environments in Online Appendix B.4.1. □

A.8 Proof of Proposition 2

Proof. Fix some environment. Suppose that the value of randomization under commitment is strictly smaller than $\kappa^*\delta$.

Let (σ^*, ρ^*) be an optimal partitional persuasion profile such that: (i) ρ^* is a pure strategy, (ii) distinct on-path messages lead to distinct actions, and (iii) any off-path message leads to an on-path action. (It is easy to see there is such an optimal persuasion profile. If Receiver were to randomize across actions in some set $\hat{A}$ following a given message, we could instead consider a strategy where Receiver deterministically takes one of Sender's preferred actions from $\hat{A}$ following that message. If Receiver were to take the same action following two distinct messages, Sender could attain the same payoff by merging those two messages.)

Let $\{\Omega_i\}_{i=1}^k$ denote the induced partition on Ω and $\{a_i\}_{i=1}^k$ denote the induced actions. Clearly, $a_i \in A^*(\Omega_i)$.

We will establish that (σ^*, ρ^*) must satisfy S-BR, which means it is a cheap-talk equilibrium. This then implies the cheap-talk payoff is weakly greater than the partitional persuasion payoff. Hence, the value of commitment is weakly smaller than the value of randomization under commitment.

Suppose toward a contradiction that (σ^*, ρ^*) does not satisfy S-BR. Then, there exists $\hat{\omega}, i, j$

such that $\hat{\omega} \in \Omega_i$ and

$$u_S(a_j, \hat{\omega}) - u_S(a_i, \hat{\omega}) > 0.$$

By the definition of δ, we have $u_S(a_j, \hat{\omega}) - u_S(a_i, \hat{\omega}) \geq \delta$.

Consider the following perturbed messaging strategy $\hat{\sigma}_\varepsilon$: for every state $\omega' \neq \hat{\omega}$, $\hat{\sigma}_\varepsilon(\omega') = \sigma^*(\omega')$; in state $\hat{\omega}$, Sender sends the original message that induces a_i with probability $1-\varepsilon$, and sends the message that induces a_j with probability ε.

We claim that for $\varepsilon = \frac{\kappa^*}{\mu_0(\hat{\omega})}$, the strategy profile $(\hat{\sigma}_\varepsilon, \rho^*)$ satisfies R-BR.[44] For any Ω_ℓ and any $a' \neq a_\ell$:

$$\sum_{\omega \in \Omega_\ell} \mu_0(\omega)[u_R(a_\ell, \omega) - u_R(a', \omega)] \geq \kappa^* \max_{a, a', \omega} |u_R(a, \omega) - u_R(a', \omega)|. \tag{17}$$

This follows from $a_\ell \in A^*(\Omega_\ell)$ and the definition of κ^*.

Relative to (σ^*, ρ^*), which of course satisfies R-BR, only two R-BR constraints are changed under $(\hat{\sigma}_\varepsilon, \rho^*)$: the message leading to a_i and the message leading to a_j. Under $\hat{\sigma}_\varepsilon$, following the message leading to action a_i, Receiver's payoff difference between taking action a_i and taking any $a' \neq a_i$ is

$$\begin{aligned}
&\sum_{\omega \in \Omega_i \setminus \{\hat{\omega}\}} \mu_0(\omega)[u_R(a_i, \omega) - u_R(a', \omega)] + (1-\varepsilon)\,\mu_0(\hat{\omega})[u_R(a_i, \hat{\omega}) - u_R(a', \hat{\omega})] \\
=& \sum_{\omega \in \Omega_i} \mu_0(\omega)[u_R(a_i, \omega) - u_R(a', \omega)] - \varepsilon \mu_0(\hat{\omega})[u_R(a_i, \hat{\omega}) - u_R(a', \hat{\omega})] \\
\geq& \sum_{\omega \in \Omega_i} \mu_0(\omega)[u_R(a_i, \omega) - u_R(a', \omega)] - \varepsilon \mu_0(\hat{\omega}) \max_{a, a', \omega} |u_R(a, \omega) - u_R(a', \omega)| \\
=& \sum_{\omega \in \Omega_i} \mu_0(\omega)[u_R(a_i, \omega) - u_R(a', \omega)] - \kappa^* \max_{a, a', \omega} |u_R(a, \omega) - u_R(a', \omega)| \\
\geq& 0,
\end{aligned}$$

where the last inequality follows from (17). Similarly, under $\hat{\sigma}_\varepsilon$, following the message leading to

[44] Note that $\varepsilon \leq 1$, because $\kappa^* \leq \kappa(\{\hat{\omega}\}) \leq \mu_0(\hat{\omega})$.

action a_j, Receiver's payoff difference between taking action a_j and taking any $a' \neq a_j$ is

$$\begin{aligned}
&\sum_{\omega\in\Omega_j} \mu_0(\omega)[u_R(a_j,\omega) - u_R(a',\omega)] + \varepsilon\mu_0(\hat{\omega})[u_R(a_j,\hat{\omega}) - u_R(a',\hat{\omega})] \\
\geq &\sum_{\omega\in\Omega_j} \mu_0(\omega)[u_R(a_j,\omega) - u_R(a',\omega)] - \varepsilon\mu_0(\hat{\omega}) \max_{a,a',\omega} |u_R(a,\omega) - u_R(a',\omega)| \\
= &\sum_{\omega\in\Omega_j} \mu_0(\omega)[u_R(a_j,\omega) - u_R(a',\omega)] - \kappa^* \max_{a,a',\omega} |u_R(a,\omega) - u_R(a',\omega)| \\
\geq &0.
\end{aligned}$$

Therefore, the strategy profile $(\hat{\sigma}_\varepsilon, \rho^*)$ satisfies R-BR.

Sender's payoff under $(\hat{\sigma}_\varepsilon, \rho^*)$ thus must be weakly lower than the persuasion payoff and weakly greater than Sender's payoff under (σ^*, ρ^*) plus $\varepsilon\mu_0(\hat{\omega})\delta$ (since $u_S(a_j,\hat{\omega}) - u_S(a_i,\hat{\omega}) \geq \delta$). Since (σ^*, ρ^*) yields the partitional persuasion payoff, this means that the value of randomization must be at least $\varepsilon\mu_0(\hat{\omega})\delta = \kappa^*\delta$, a contradiction. □

B Online Appendix

B.1 Infinite games

In this section, we illustrate some of the issues that arise when A is uncountable.[45] We also consider how the intuitions behind our results translate to infinite games that are limits of sequences of finite games. Recall that in Section 7.2, we already observed that an atomless prior implies no value for randomization.

B.1.1 Uncountable A

Consider a binary-state version of the quadratic-loss, constant-bias preferences from Crawford and Sobel (1982). Suppose $\Omega = \{0,1\}$ and let μ_0 be equiprobable. We will contrast the case where $A = [0,1]$ and the case where $A = \left\{0, \frac{1}{n}, \frac{2}{n}, ..., \frac{n-1}{n}, 1\right\}$ for some even integer $n \geq 2$. We refer to the former as the *interval case* and the latter as the *finite case*.

Receiver's utility is $u_R(a,\omega) = -(a-\omega)^2$. Sender's utility is $u_S(a,\omega) = -(a-\omega-b)^2$ for

[45] We do not explore the specific case of countably infinite A.

some $b > 0$. We say a profile (σ, ρ) is *full revelation* if there exist disjoint subsets $M_0, M_1 \subseteq M$ such that $\sigma(M_0|\omega = 0) = 1$, $\sigma(M_1|\omega = 1) = 1$.

The parameter region where $b > \frac{1}{2}$ illustrates the contrast between the interval case and the finite case.[46] In the interval case, only a full revelation profile yields the persuasion payoff, but such a profile cannot be a cheap-talk equilibrium.[47] Hence, in the interval case, in contrast to Theorem 1, commitment has value even though committed Sender does not value randomization. In the finite case, however, the assumptions underlying Theorem 1 apply. In this case, commitment is valuable and committed Sender values randomization.[48]

B.1.2 Limits of finite games

A continuous communication game is a tuple $\Gamma := (\Omega, A, M, u_S, u_R, \mu_0)$ where the state space Ω, message space M, and action space A are compact subsets of Euclidean space with $|M| \geq \max\{|\Omega|, |A|\}$, $u_S, u_R : A \times \Omega \to \mathbb{R}$ are continuous, and $\mu_0 \in \Delta(\Omega)$ is the prior.[49] Note that if Ω and A are finite, the game is trivially continuous.

A discretization of Γ is a game $\mathcal{G} = (\hat{\Omega}, \hat{A}, \hat{M}, \hat{u}_S, \hat{u}_R, \hat{\mu}_0)$ where $\hat{\Omega} \subseteq \Omega$, $\hat{M} \subseteq M$, $\hat{A} \subseteq A$ are finite subsets with $|\hat{M}| \geq \max\{|\hat{\Omega}|, |\hat{A}|\}$, $\hat{u}_S$ and $\hat{u}_R$ are the restrictions of u_S and u_R to $\hat{A} \times \hat{\Omega}$, and $\hat{\mu}_0 \in \Delta\Omega$ with support $\text{Supp}(\hat{\mu}_0) = \hat{\Omega}$. We say a discretization $\mathcal{G}$ satisfies partitional-unique-response if $(\hat{u}_S, \hat{u}_R)$ satisfies partitional-unique-response.

A sequence of discretizations $\{\mathcal{G}^n\}_{n=1}^{\infty}$, where $\mathcal{G}^n = (\Omega_n, A_n, M_n, u_S^n, u_R^n, \mu_0^n)$, converges to Γ if $d_H^{\Omega}(\Omega_n, \Omega) \to 0$, $d_H^A(A_n, A) \to 0$, and $\mu_0^n \to \mu_0$ weakly in $\Delta(\Omega)$.[50] We then say that Γ is the limit of $\{\mathcal{G}^n\}_{n=1}^{\infty}$.

For any game (continuous or discretization) $\mathcal{G}$, let $V_{BP}(\mathcal{G})$ denote the persuasion payoff and

[46] When $b \leq \frac{1}{2}$, it does not matter whether we consider the interval or the finite case. In both cases, full revelation yields the persuasion payoff and is a cheap-talk equilibrium. Thus, neither commitment nor randomization is valuable.

[47] Due to the strict convexity of Sender's indirect utility function, the persuasion payoff can only be achieved by full revelation. However, a full revelation profile cannot be a cheap-talk equilibrium: type $\omega = 0$ would deviate and send a message in M_1, because $u_S(0, 0) = -b^2 < -(1 - b)^2 = u_S(1, 0)$.

[48] To see that committed Sender values randomization, note that a full revelation profile yields the payoff $-[\mu_0(\omega = 1)(0 - 0 - b)^2 + (1 - \mu_0(\omega = 1))(1 - 1 - b)^2] = -b^2$. Providing no information yields the payoff $-[\mu_0(\omega = 1)(1/2 - 0 - b)^2 + (1 - \mu_0(\omega = 1))(1/2 - 1 - b)^2] = -(b^2 + 1/4)$. Therefore, Sender's partitional persuasion payoff is $-b^2$. But, Sender can obtain a strictly higher payoff by inducing posteriors $\mu = \frac{1}{2n}$ and $\mu = 1$. When $\mu = \frac{1}{2n}$, Receiver's optimal action is $a = \frac{1}{n}$, so Sender's interim value is $-[(1 - \frac{1}{2n})(\frac{1}{n} - 0 - b)^2 + (\frac{1}{2n})(\frac{1}{n} - 1 - b)^2] = -b^2 + \frac{2b-1}{2n} > -b^2$. When $\mu = 1$, Sender's interim value is $-b^2$. Sender's ex-ante payoff is a convex combination of the two interim values, which is strictly higher than $-b^2$.

[49] We use $\Delta(\Omega)$ to denote the set of Borel probability measures on Ω, endowed with the weak topology.

[50] We use d_H^{Ω} and d_H^A to denote the Hausdorff distance on Ω and A respectively.

$V_{CT}(\mathcal{G})$ denote the cheap-talk payoff.

Proposition 4. *Consider a continuous communication game Γ such that A is convex and u_R is strictly concave in a. Suppose Γ is the limit of $\{\mathcal{G}^n\}_{n=1}^{\infty}$, each $\mathcal{G}^n$ satisfies partitional-unique-response, and in each $\mathcal{G}^n$, committed Sender does not value randomization. Then, Sender does not value commitment in Γ.*

We begin the proof with two lemmas.

Lemma 12. $\lim_{n\to\infty} V_{BP}(\mathcal{G}^n) = V_{BP}(\Gamma)$.

Proof. For any belief $\mu \in \Delta(\Omega)$, let

$$a^*(\mu) := \arg\max_{a\in A} \int u_R(a,\omega)d\mu(\omega)$$

denote Receiver's best response in the continuous game Γ. Since u_R is strictly concave and A is compact, $a^*(\mu)$ is unique and continuous in μ. Therefore, Sender's indirect utility function

$$v(\mu) := \int u_S(a^*(\mu),\omega)d\mu(\omega)$$

is also continuous.

For each $\mathcal{G}^n$, let

$$A_n^*(\mu) := \arg\max_{a\in A_n} \int u_R(a,\omega)d\mu(\omega)$$

denote Receiver's best responses in the finite game $\mathcal{G}^n$. Define Sender's indirect utility function in the finite game as[51]

$$v_n(\mu) := \max_{a\in A_n^*(\mu)} \int u_S(a,\omega)d\mu(\omega).$$

We first prove that

$$\sup_{\mu\in\Delta(\Omega)} \sup_{a\in A_n^*(\mu)} \|a - a^*(\mu)\| \longrightarrow 0$$

as $n \to \infty$.

Fix $\varepsilon > 0$. Define $D_\varepsilon := \{(\mu,a) \in \Delta(\Omega)\times A : \|a - a^*(\mu)\| \geq \varepsilon\}$, which is compact. By the

[51]Only the restriction of v_n to $\Delta(\Omega_n)$ is payoff-relevant for $\mathcal{G}^n$, but it is useful to define v_n on the entire $\Delta(\Omega)$.

uniqueness of $a^*(\mu)$,

$$\int u_R(a^*(\mu),\omega)d\mu(\omega) - \int u_R(a,\omega)d\mu(\omega) > 0$$

for all $(\mu, a) \in D_\varepsilon$. Since D_ε is compact and the value difference is continuous in (μ, a), there exists $\delta_\varepsilon > 0$ such that

$$\int u_R(a^*(\mu),\omega)d\mu(\omega) - \int u_R(a,\omega)d\mu(\omega) \geq \delta_\varepsilon \tag{18}$$

for all $(\mu, a) \in D_\varepsilon$.

For each $\mu \in \Delta(\Omega)$, choose $a_n(\mu) \in A_n$ such that

$$\|a_n(\mu) - a^*(\mu)\| \leq d_H^A(A_n, A).$$

By the continuity of u_R on compact $A \times \Omega$, for sufficiently large n,

$$\int u_R(a_n(\mu),\omega)d\mu(\omega) \geq \int u_R(a^*(\mu),\omega)d\mu(\omega) - \frac{\delta_\varepsilon}{2} \tag{19}$$

for all $\mu \in \Delta(\Omega)$.

Now suppose toward a contradiction that for some large n, some μ, and some $a_n^* \in A_n^*(\mu)$,

$$\|a_n^* - a^*(\mu)\| \geq \varepsilon.$$

By (18),

$$\int u_R(a^*(\mu),\omega)d\mu(\omega) - \int u_R(a_n^*,\omega)d\mu(\omega) \geq \delta_\varepsilon.$$

Combined with (19),

$$\int u_R(a_n(\mu),\omega)d\mu(\omega) \geq \int u_R(a_n^*,\omega)d\mu(\omega) + \frac{\delta_\varepsilon}{2} > \int u_R(a_n^*,\omega)d\mu(\omega),$$

which contradicts $a_n^* \in A_n^*(\mu)$. Therefore,

$$\sup_{\mu\in\Delta(\Omega)} \sup_{a\in A_n^*(\mu)} \|a - a^*(\mu)\| \longrightarrow 0 \tag{20}$$

as $n \to \infty$.

We next show that $\sup_{\mu\in\Delta(\Omega)} |v(\mu) - v_n(\mu)| \to 0$. Observe that

$$\begin{aligned} |v(\mu) - v_n(\mu)| &= \left| \int u_S(a^*(\mu), \omega) d\mu(\omega) - \max_{a \in A_n^*(\mu)} \int u_S(a, \omega) d\mu(\omega) \right| \\ &= \left| \int u_S(a^*(\mu), \omega) d\mu(\omega) - \int u_S(\tilde{a}_n, \omega) d\mu(\omega) \right| \end{aligned}$$

for any $\tilde{a}_n \in A_n^*(\mu)$ that maximizes Sender's payoff. By the continuity of u_S and (20), we have

$$\sup_{\mu\in\Delta(\Omega)} |v(\mu) - v_n(\mu)| \to 0. \tag{21}$$

By Kamenica and Gentzkow (2011) and its online appendix (Proposition 3), $V_{BP}(\Gamma) = cav(v)(\mu_0)$ and $V_{BP}(\mathcal{G}^n) = cav(v_n)(\mu_0^n)$.

Note that for any $\tau \in \Delta(\Delta(\Omega))$,

$$\left| \int v_n(\mu) d\tau(\mu) - \int v(\mu) d\tau(\mu) \right| = \left| \int \big(v_n(\mu) - v(\mu)\big) d\tau(\mu) \right| \leq \sup_{\mu\in\Delta(\Omega)} |v(\mu) - v_n(\mu)|.$$

Therefore, for every $\mu \in \Delta(\Omega)$,

$$|cav(v_n)(\mu) - cav(v)(\mu)| \leq \sup_{\mu\in\Delta(\Omega)} |v(\mu) - v_n(\mu)|.$$

Together with (21), we have

$$|cav(v_n)(\mu) - cav(v)(\mu)| \to 0. \tag{22}$$

In addition, since v is continuous, $cav(v)$ is also continuous. Since μ_0^n converges to μ_0 weakly,

$$|cav(v)(\mu_0^n) - cav(v)(\mu_0)| \to 0. \tag{23}$$

Therefore,

$$\begin{aligned}|V_{BP}(\mathcal{G}^n) - V_{BP}(\Gamma)| &= |cav(v_n)(\mu_0^n) - cav(v)(\mu_0)| \\ &\leq |cav(v_n)(\mu_0^n) - cav(v)(\mu_0^n)| + |cav(v)(\mu_0^n) - cav(v)(\mu_0)| \\ &\to 0 \qquad \text{by (22) and (23).}\end{aligned}$$

□

Lemma 13. $\limsup_{n\to\infty} V_{CT}(\mathcal{G}^n) \leq V_{CT}(\Gamma)$.

Proof. For each $\mathcal{G}^n$, since the committed Sender does not value randomization and the environment satisfies partitional-unique-response, there exists a pure-strategy cheap-talk equilibrium (σ_n, ρ_n) attaining $V_{CT}(\mathcal{G}^n) = V_{BP}(\mathcal{G}^n)$.

For each $\omega \in \Omega_n$, let $f_n(\omega) := \rho_n(\sigma_n(\omega))$ denote the action induced in state ω. Let $\pi^n \in \Delta(A \times \Omega)$, defined as $\pi^n := \sum_{\omega\in\Omega_n} \mu_0^n(\omega)\delta_{(f_n(\omega),\omega)}$, be the induced state-action distribution.

Since π^n is induced by a pure-strategy cheap-talk equilibrium, it satisfies

(i) $\pi^n_\Omega = \mu_0^n$;

(ii) for any $(a, \omega) \in \text{Supp}(\pi^n)$

$$u_S(a, \omega) \geq u_S(a', \omega), \quad \forall a' \in \text{Supp}(\pi^n_A). \tag{finite-S-BR}$$

(iii) for any $a \in A_n$,

$$\sum_{\omega\in\Omega_n} \pi^n(a, \omega)[u_R(a, \omega) - u_R(a', \omega)] \geq 0, \quad \forall a' \in A_n. \tag{finite-R-BR}$$

Since $\Delta(A \times \Omega)$ is compact, there exists a convergent subsequence of π^n. Let π^{n_k} be any such subsequence and π^* be its limit. We will show that π^* can be induced in a cheap-talk equilibrium in Γ. Since u_S is continuous and bounded, $\lim_{k\to\infty} V_{CT}(\mathcal{G}^{n_k}) = \lim_{k\to\infty} \int u_S(a, \omega)\, d\pi^{n_k}(a, \omega) = \int u_S(a, \omega)\, d\pi^*(a, \omega)$. This then implies $V_{CT}(\Gamma) \geq \limsup_{n\to\infty} V_{CT}(\mathcal{G}^n)$.

To show that π^* can be induced in a cheap-talk equilibrium, we establish that for any $(a, \omega) \in \text{Supp}(\pi^*)$,

$$u_S(a, \omega) \geq u_S(a', \omega), \quad \forall a' \in \text{Supp}(\pi^*_A), \tag{S-BR}$$

and for any measurable deviation function $g : A \to A$,

$$\int [u_R(a, \omega) - u_R(g(a), \omega)] \, d\pi^*(a, \omega) \geq 0. \tag{R-BR}$$

Then, by the standard "direct recommendation" argument, one can construct a cheap-talk equilibrium that yields π^*.

We start with S-BR. Let $(a, \omega) \in \text{Supp}(\pi^*)$ and take any alternative $a' \in \text{Supp}(\pi^*_A)$. Since the support of π^*_A is contained in the limit of $\text{Supp}(\pi^{n_k}_A)$, there exist sequences $(a_{n_k}, \omega_{n_k}) \in \text{Supp}(\pi^{n_k})$ converging to (a, ω) and $a'_{n_k} \in \text{Supp}(\pi^{n_k}_A)$ converging to a'.

For any $\mathcal{G}^{n_k}$, finite-S-BR implies

$$u_S(a_{n_k}, \omega_{n_k}) \geq u_S(a'_{n_k}, \omega_{n_k}).$$

Since u_S is continuous, taking $k \to \infty$ yields $u_S(a, \omega) \geq u_S(a', \omega)$.

We next establish R-BR. Fix $a' \in A$. Since $A_{n_k} \to A$, there exists a sequence of $a'_{n_k} \in A_{n_k}$ such that $a'_{n_k} \to a'$.

For each $\mathcal{G}^{n_k}$, finite-R-BR implies that, for any continuous test function $t : A \to \mathbb{R}_+$,

$$\int [u_R(a, \omega) - u_R(a'_{n_k}, \omega)] t(a) \, d\pi^{n_k}(a, \omega) \geq 0.$$

Since u_R is continuous and $a'_{n_k} \to a'$, the integrands converge uniformly to $[u_R(a, \omega) - u_R(a', \omega)]t(a)$. Since π^{n_k} converges to π^* weakly, we have

$$\int [u_R(a, \omega) - u_R(a', \omega)] t(a) \, d\pi^*(a, \omega) \geq 0.$$

Define a signed measure $\xi_{a'}$ on A by

$$\xi_{a'}(E) := \int_{E\times\Omega} [u_R(a,\omega) - u_R(a',\omega)] d\pi^*(a,\omega).$$

From the previous inequality,

$$\int_A t(a) d\xi_{a'}(a) \geq 0$$

for all continuous $t : A \to \mathbb{R}_+$. Hence, $\xi_{a'}$ is a positive measure. Therefore, for every Borel set $E \subseteq A$,

$$\int_{E\times\Omega} [u_R(a,\omega) - u_R(a',\omega)] d\pi^*(a,\omega) \geq 0.$$

Now take a countable dense subset $D \subseteq A$. The inequality holds for every $a' \in D$ and every Borel $E \subseteq A$.

Let $g : A \to D$ be a finite-valued measurable deviation rule. Let $\{a_1, ..., a_N\}$ denote the image of g. Define $E_i = g^{-1}(a_i)$, so $\{E_i\}_{i=1}^N$ is a finite measurable partition. Then

$$\begin{aligned}
\int u_R(a,\omega) d\pi^* &= \sum_i \int_{E_i\times\Omega} u_R(a,\omega) d\pi^* \\
&\geq \sum_i \int_{E_i\times\Omega} u_R(a_i,\omega) d\pi^* \\
&= \int u_R(g(a),\omega) d\pi^*.
\end{aligned}$$

Finally, since A is compact metric, every measurable $g : A \to A$ can be uniformly approximated by measurable finite-valued maps $g_m : A \to D$. Since u_R is uniformly continuous, we have $u_R(g_m(a),\omega) \to u_R(g(a),\omega)$ uniformly in (a,ω). Passing to the limit yields

$$\int u_R(a,\omega) d\pi^* \geq \int u_R(g(a),\omega) d\pi^*.$$

□

Lemmas 12 and 13, together with Theorem 1, imply that $V_{BP}(\Gamma) = \lim_{n\to\infty} V_{BP}(\mathcal{G}^n) = \limsup_{n\to\infty} V_{CT}(\mathcal{G}^n) \leq V_{CT}(\Gamma)$. Since $V_{BP}(\Gamma) \geq V_{CT}(\Gamma)$ holds trivially, we obtain $V_{BP}(\Gamma) = V_{CT}(\Gamma)$.

B.2 State-independent preferences

As we mentioned in the discussion of related literature, several papers examine value of commitment under the assumption that Sender has state-independent preferences. To connect to that literature, it is worth asking whether our results hold under that assumption. When $|A| \geq 3$, state-independent preferences by Sender mean that we are not in a scant-indifferences environment, so the proofs above do not apply. Nonetheless, Theorems 1 and 2 indeed remain true.

To formalize this, say that environment (u_S, u_R) is transparent if there exists some function $\phi : A \to \mathbb{R}$ such that $u_S(a, \omega) = u_S(a, \omega') \equiv \phi(a)$ for any a, ω, ω'. The set of all transparent environments is $[0,1]^{|A|(|\Omega|+1)}$. A set of transparent environments is *transparently-generic* if it has Lebesgue measure one in $\mathbb{R}^{|A|(|\Omega|+1)}$. We say that a claim holds *generically in transparent environments*, if it holds for a transparently-generic set of environments.

A transparent environment (u_S, u_R) satisfies *no-duplicate-actions* if for any $a \neq a'$, $\phi(a) \neq \phi(a')$. Clearly, the set of no-duplicate-actions transparent environments is transparently-generic.

A strategy profile (σ, ρ) is a *simple babbling cheap-talk equilibrium* if it is a cheap-talk equilibrium in which $|M_\sigma| = 1$ and $\rho(m) = a_0$ for all $m \in M$ and for some $a_0 \in A$.

Lemma 14. *In a no-duplicate-actions transparent environment, if commitment has no value, then there exists a simple babbling cheap-talk equilibrium that yields the persuasion payoff.*

Proof. If commitment has no value, some cheap-talk equilibrium, denoted by (σ, ρ), yields the persuasion payoff. First, we claim that ρ must be pure-on-path. Suppose by contradiction that at some on-path message m, $|\operatorname{Supp}(\rho(\cdot|m))| > 1$. R-BR then implies that Receiver must be indifferent among all actions in $\operatorname{Supp}(\rho(\cdot|m))$. Since the environment satisfies no-duplicate-actions, Sender must strictly prefer one of the actions in $\operatorname{Supp}(\rho(\cdot|m))$ over all others. Therefore, an alternative strategy profile where Receiver breaks ties in favor of Sender would still satisfy R-BR while strictly improving Sender's payoff.

Since (σ, ρ) is a cheap-talk equilibrium and the environment is transparent, S-BR implies that for any $m, m' \in M_\sigma$, we have $\phi(\rho(m)) = \phi(\rho(m'))$. Moreover, because the environment satisfies no-duplicate-actions, it follows that $\rho(m) = \rho(m')$ for any $m, m' \in M_\sigma$; that is, only a single action is induced in equilibrium.

We now construct a simple babbling cheap-talk equilibrium $(\hat{\sigma}, \hat{\rho})$ that yields the same payoff as (σ, ρ). Choose an arbitrary message $m_0 \in M_\sigma$. Let $\hat{\sigma}(m_0|\omega) = 1$ for all $\omega \in \Omega$, and $\hat{\rho}(m) = \rho(m_0)$ for all $m \in M$. The strategy profile $(\hat{\sigma}, \hat{\rho})$ trivially satisfies S-BR and yields the same payoff as (σ, ρ). Since (σ, ρ) satisfies R-BR, it follows that for each $m \in M_\sigma$, $\sum_\omega \mu_0(\omega)\sigma(m|\omega)u_R(\rho(m), \omega) \geq \sum_\omega \mu_0(\omega)\sigma(m|\omega)u_R(a, \omega)$ for all $a \in A$. Summing over all m, we obtain

$$\sum_{\omega\in\Omega} \mu_0(\omega) \sum_{m\in M} \sigma(m|\omega)u_R(\rho(m), \omega) \geq \sum_{\omega\in\Omega} \mu_0(\omega) \sum_{m\in M} \sigma(m|\omega)u_R(a, \omega) \quad \text{for all } a \in A.$$

Since $\rho(m) = \rho(m_0)$ for all $m \in M_\sigma$, we can rewrite the inequality as

$$\sum_{\omega\in\Omega} \mu_0(\omega)u_R(\rho(m_0), \omega) \geq \sum_{\omega\in\Omega} \mu_0(\omega)u_R(a, \omega) \quad \text{for all } a \in A,$$

which then implies $(\hat{\sigma}, \hat{\rho})$ satisfies R-BR. Therefore, $(\hat{\sigma}, \hat{\rho})$ is a simple babbling cheap-talk equilibrium that yields the persuasion payoff. □

Theorem 4. *Generically in transparent environments, commitment is valuable if and only if committed Sender values randomization.*

Proof. We establish the equivalence for any transparent environment that satisfies both partitional-unique-response and no-duplicate-actions. Recall that whether an environment satisfies partitional-unique-response does not depend on Sender's preferences, so the same argument as in Lemma 1 implies that the set of partitional-unique-response transparent environments is transparently-generic. Moreover, the set of no-duplicate-actions transparent environments is transparently-generic. Therefore, the set of transparent environments that satisfy both properties is also transparently-generic.

The same arguments that establish (i) implies (iii) and (iii) implies (ii) in Theorem 1' apply directly to any partitional-unique-response transparent environment, thereby proving the "only if" direction. The "if" direction follows immediately from Lemma 14. □

Theorem 5. *Generically in transparent environments, commitment is valuable if cheap-talk Sender values randomization.*

Proof. The theorem follows immediately from Lemma 14 and the fact that the set of no-duplicate-actions transparent environments is transparently-generic. □

An alternative way of seeing Theorem 5 is through the quasi-concavification characterization of the cheap-talk payoff (Lipnowski and Ravid 2020), and the concavification characterization of the persuasion payoff (Kamenica and Gentzkow 2011). Consider a no-duplicate-actions transparent environment. For each $\mu \in \Delta\Omega$, let $A_R^*(\mu) := \arg\max_{a \in A} \sum_\omega u_R(a,\omega)\mu(\omega)$ denote the set of Receiver-optimal actions under belief μ. Let $v(\mu) = \max_{a \in A_R^*(\mu)} \sum_\omega u_S(a,\omega)\mu(\omega)$ denote Sender's indirect utility function. Let $\bar{v}$ denote the quasi-concavification of v and $\hat{v}$ denote the concavification of v. Let V^{CT} denote the cheap-talk payoff, and V^{BP} denote the persuasion payoff. Lipnowski and Ravid (2020) establish that $V^{CT} = \bar{v}(\mu_0)$ and Kamenica and Gentzkow (2011) establish that $V^{BP} = \hat{v}(\mu_0)$.

By the definition of quasi-concavification and Carathéodory's theorem, there exists $1 \leq k \leq |\Omega|$, $\mu_1, ..., \mu_k \in \Delta(\Omega)$ and $\lambda_1, ..., \lambda_k > 0$ such that $\mu_0 = \sum_{i=1}^k \lambda_i \mu_i$, $\sum_{i=1}^k \lambda_i = 1$, and $v(\mu_i) \geq V^{CT}$ for $i = 1, ..., k$.

If randomization is valuable in cheap talk, $V^{CT} = \bar{v}(\mu_0) > v(\mu_0)$, because $v(\mu_0)$ can be achieved in a babbling equilibrium. We claim that $v(\mu_i) > \bar{v}(\mu_0)$ for some i. Suppose, toward a contradiction, that $v(\mu_i) = \bar{v}(\mu_0)$ for all $i = 1, ..., k$. Since the environment satisfies no-duplicate-actions, there exists a unique action $a^* \in A$ such that $\phi(a^*) = \bar{v}(\mu_0)$. This means that $a^* \in A_R^*(\mu_i)$ for all i. Since the set of posteriors μ such that $a^* \in A_R^*(\mu)$ is convex, we have $a^* \in A_R^*(\mu_0)$. This implies $v(\mu_0) \geq \bar{v}(\mu_0)$, a contradiction.

Since there exists i such that $v(\mu_i) > V^{CT}$,

$$V^{BP} = \hat{v}(\mu_0) \geq \sum_{i=1}^k \lambda_i v(\mu_i) > V^{CT}.$$

Therefore, commitment is valuable.

Figure 3 illustrates this geometric proof. In the figure, the black curve depicts Sender's indirect utility function in a no-duplicate-actions transparent environment with three actions and two states. The red curve is its quasi-concave envelope. The cheap-talk Sender values randomization and $\bar{v}(\mu_0) > v(\mu_0)$. This means that there exist $\lambda \in (0,1)$ and $\mu_1, \mu_2 \in \Delta(\Omega)$ such that $\lambda\mu_1 + (1-\lambda)\mu_2 = \mu_0$, $v(\mu_1) \geq \bar{v}(\mu_0)$ and $v(\mu_2) \geq \bar{v}(\mu_0)$. In addition, at least one of the two inequalities must be strict, and in this example it is $v(\mu_2) > \bar{v}(\mu_0)$. Hence, the committed Sender can induce posteriors

μ_1, μ_2 and achieve a payoff of $\lambda v(\mu_1) + (1-\lambda)v(\mu_2) > \bar{v}(\mu_0)$, depicted by the blue curve. Therefore, commitment is valuable.

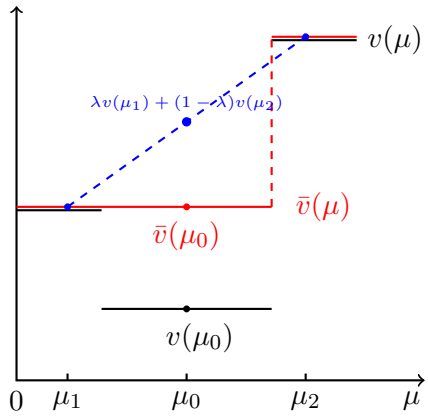


Figure 3

B.3 Strictly positive value of commitment and randomization

Theorem 1 tells us that, generically, commitment has zero value if and only if randomization has zero value. A natural question is whether, generically, a small value of commitment implies or is implied by a small value of randomization. Here we demonstrate that the answer is no.

We show that there is a $\Delta > 0$ such that for any $\varepsilon > 0$: (i) there is a positive measure of environments where the value of commitment is greater than Δ and the value of randomization is less than ε, and (ii) there is a positive measure of environments where the value of randomization is greater than Δ and the value of commitment is less than ε.[52]

We begin by illustrating the role of the genericity condition for Theorem 1. We present two examples. The first example presents an environment (that violates partitional-unique-response) where commitment is valuable but randomization is not. The second example presents an environment (that violates scant indifferences) where randomization is valuable but commitment is not.

[52] Recall that we assume that the range of the utility functions is $[0, 1]$ in order to succinctly state our results about how often commitment is valuable. If we relax the assumption that the range of the utility functions is uniformly bounded, we can alternatively show that *for any* $\Delta > 0$ and any $\varepsilon > 0$: (i) there is a positive measure of environments where the value of commitment is greater than Δ and the value of randomization is less than ε, and (ii) there is a positive measure of environments where the value of randomization is greater than Δ and the value of commitment is less than ε.

Then, we build on the first example to show how to construct, given any ε, a positive measure of environments where the value of commitment is above $\frac{1}{6}$ but the value of randomization is less than ε. We build on the second example to show how to construct, given any ε, a positive measure of environments where the value of randomization is above $\frac{1}{13}$ but the value of commitment is less than ε.

Example 3. Consider $\Omega = \{\omega_1, \omega_2\}$ with prior $\mu_0(\omega_1) = \mu_0(\omega_2) = 0.5$ and $A = \{a_1, a_2\}$. Players' payoffs are given in Table 2. Receiver's best response is a_1 when $\mu := \mu(\omega_2) \in [0, 1)$, and she is indifferent between a_1 and a_2 when $\mu = 1$. The concavification of Sender's indirect utility function is depicted in Figure 4.

u_S	a_1	a_2
ω_1	0	2/3
ω_2	0	2/3

u_R	a_1	a_2
ω_1	1/2	0
ω_2	1/2	1/2

Table 2: Sender's and Receiver's payoffs

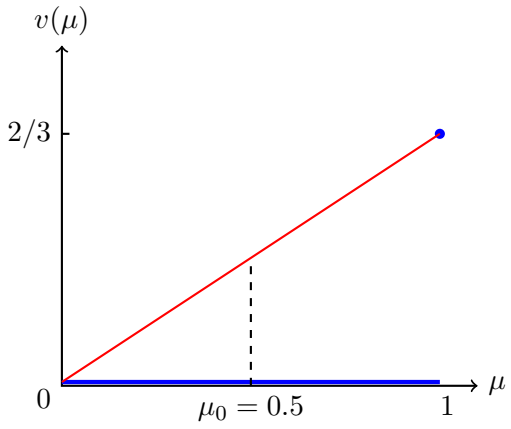


Figure 4: Concavification

Clearly, full revelation is the unique optimal information structure, which yields a payoff of 1/3; therefore, Sender does not value randomization. In addition, the only possible cheap-talk equilibrium outcome is babbling, which yields a payoff of 0. Hence, commitment is valuable.

Example 4. Consider $\Omega = \{\omega_1, \omega_2\}$ with prior $\mu_0(\omega_1) = \mu_0(\omega_2) = 0.5$ and $A = \{a_1, a_2, a_3\}$. Players' payoffs are given in Table 3. Receiver's best response is a_1 when $\mu \in [0, 1/3]$, a_2 when $\mu \in [1/3, 2/3]$, and a_3 when $\mu := \mu(\omega_2) \in [2/3, 1]$. This leads to Sender's indirect utility function (blue) and its concave envelope (red) depicted in Figure 5.

Sender values randomization, because the unique optimal information structure that induces a posterior 1/3 cannot be generated by a partitional messaging strategy. Since Sender's indirect

u_S	a_1	a_2	a_3
ω_1	1/3	2/3	0
ω_2	5/6	1/6	1/2

u_R	a_1	a_2	a_3
ω_1	3/4	1/2	0
ω_2	0	1/2	3/4

Table 3: Sender's and Receiver's payoffs

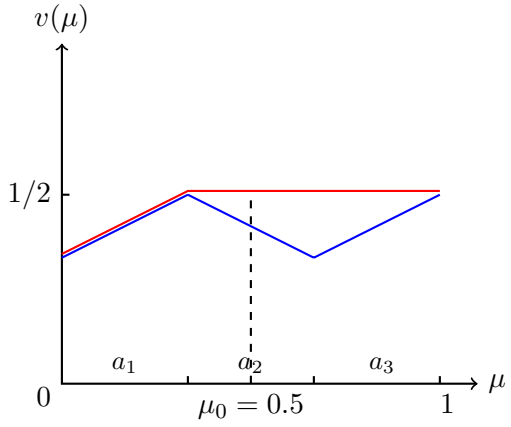


Figure 5: Concavification

utility function is continuous, by Lipnowski (2020), Sender does not value commitment. For the sake of completeness, we construct a cheap-talk equilibrium that yields the persuasion payoff 1/2 to Sender.

Consider a strategy profile (σ, ρ) with two on-path messages m_1, m_2: $\sigma(m_1|\omega_2) = 1/2$, $\sigma(m_2|\omega_2) = 1/2$, $\sigma(m_1|\omega_1) = 1$; $\rho(a_1|m_1) = 1/2$, $\rho(a_2|m_1) = 1/2$, $\rho(a_3|m_2) = 1$.

The profile satisfies R-BR because the posterior upon observing m_1 is 1/3 and when observing m_2 is 1. We next show that the profile also satisfies S-BR. For type ω_2 Sender, the expected payoff of sending message m_2 is 1/2 and the expected payoff of sending message m_1 is $\frac{1}{2} \cdot \frac{5}{6} + \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{2}$, so type ω_2 Sender is indifferent and has no incentive to deviate. For type ω_1 Sender, the expected payoff of sending message m_2 is 0 and the expected payoff of sending message m_1 is $\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{2}$, so he strictly prefers sending message m_1. Hence, (σ, ρ) is a cheap-talk equilibrium that yields the persuasion payoff.

B.3.1 Small value of randomization does not imply small value of commitment

We construct a positive measure of environments where the value of commitment is bounded away from zero but the value of randomization is arbitrarily small. Formally, given an environment (u_S, u_R), let $\Delta_{\mathcal{R}}(u_S, u_R) = \textit{Persuasion Payoff} - \textit{Partitional Persuasion Payoff}$ and $\Delta_{\mathcal{C}}(u_S, u_R) = \textit{Persuasion Payoff} - \textit{Cheap-Talk Payoff}$ denote the value of randomization and the value of com-

u_S	a_1	a_2
ω_1	$0 + s_{11}$	$2/3 + s_{12}$
ω_2	$0 + s_{21}$	$2/3 + s_{22}$

u_R	a_1	a_2
ω_1	$1/2 + r_{11}$	r_{12}
ω_2	$1/2 + r_{21}$	$1/2 + r_{22}$

Table 4: Sender's and Receiver's payoffs

mitment, respectively. We will establish that for any $\varepsilon > 0$, there exists a positive measure set of environments E such that for any $(u_S, u_R) \in E$, we have $\Delta_{\mathcal{R}}(u_S, u_R) < \varepsilon$ and $\Delta_{\mathcal{C}}(u_S, u_R) > 1/6$.

Fix any $\varepsilon > 0$. We perturb players' payoffs in Example 3 to construct the desired positive share of environments. The idea is that for sufficiently small perturbations in an appropriate direction, the changes to the persuasion payoff, partitional persuasion payoff, and cheap-talk payoff will also be small. Since in Example 3, the value of randomization is zero and the value of commitment is positive, we obtain a positive measure of environments with a small value of randomization and a value of commitment bounded away from zero.

Players' payoffs are as in Table 4, where s_{ij}, r_{ij} are the perturbations to Sender's and Receiver's payoffs, respectively, when action a_j is taken in state ω_i.

Let $s_{ij} \in [0, \delta]$, $r_{11}, r_{21} \in [-\delta, 0]$, and $r_{12}, r_{22} \in [0, \delta]$, where $\delta > 0$. These perturbations generate a positive measure set of environments, denoted by E^δ. We will establish that if $\delta < \min\{\frac{1}{8}, \frac{\varepsilon}{4(1+\varepsilon)}\}$, then for any $(u_S, u_R) \in E^\delta$, the value of commitment is greater than $1/6$ and the value of randomization is less than ε.

Consider any $(u_S, u_R) \in E^\delta$. Since $\delta < 1/8$, Receiver's best response is a_2 iff $\mu \geq \mu^* := \frac{1/2 + r_{11} - r_{12}}{1/2 + r_{11} - r_{21} - r_{12} + r_{22}} \in [1 - 4\delta, 1]$. Sender's indirect utility function is therefore

$$v(\mu) = \begin{cases} s_{11} + (s_{21} - s_{11})\mu & \text{if } \mu < \mu^* \\ \frac{2}{3} + s_{12} + (s_{22} - s_{12})\mu & \text{if } \mu \geq \mu^*. \end{cases}$$

By inducing beliefs $\mu = 0$ and $\mu = \mu^*$, Sender achieves his persuasion payoff

$$\frac{1}{2\mu^*}[\frac{2}{3} + s_{12} + (s_{22} - s_{12})\mu^*] + (1 - \frac{1}{2\mu^*})s_{11}.$$

Meanwhile, full revelation yields a payoff of

$$\frac{1}{3} + \frac{s_{11} + s_{22}}{2}.$$

Therefore,

$$\begin{aligned}
\Delta_{\mathcal{R}}(u_S, u_R) \leq & \frac{1}{2\mu^*}[\frac{2}{3} + s_{12} + (s_{22} - s_{12})\mu^*] + (1 - \frac{1}{2\mu^*})s_{11} - \frac{1}{3} - \frac{s_{11} + s_{22}}{2} \\
= & \frac{1 - \mu^*}{\mu^*}(\frac{1}{3} + \frac{s_{12} - s_{11}}{2}) \\
\leq & \frac{4\delta}{1 - 4\delta}(\frac{1}{3} + \frac{\delta}{2}) \\
< & \varepsilon
\end{aligned}$$

where the third line follows from $\mu^* \geq 1 - 4\delta$ and $s_{ij} \in [0, \delta]$, and the last line follows from $\delta < \frac{\varepsilon}{4(1+\varepsilon)}$ and $\frac{1}{3} + \frac{\delta}{2} < 1$.

In addition, since $\delta < 1/8$, a_2 is Sender's preferred action regardless of the state. It follows that any cheap-talk equilibrium outcome must be the babbling outcome where Receiver takes action a_1 with probability 1. Therefore, Sender's cheap-talk payoff is $\frac{s_{11}+s_{21}}{2}$. Hence, $\Delta_{\mathcal{C}}(u_S, u_R) =$ *Persuasion Payoff − Cheap-Talk Payoff* $\geq \frac{1}{3} + \frac{s_{11}+s_{22}}{2} - \frac{s_{11}+s_{21}}{2} > \frac{1}{6}$, where the weak inequality follows from the fact that the persuasion payoff is greater than the payoff from full revelation, and the strict inequality follows from $\frac{s_{22}-s_{21}}{2} \geq \frac{-\delta}{2} > -\frac{1}{6}$.

B.3.2 Small value of commitment does not imply small value of randomization

We will establish that for any $\varepsilon > 0$, there exists a positive measure set of environments E such that for any $(u_S, u_R) \in E$, we have $\Delta_{\mathcal{C}}(u_S, u_R) < \varepsilon$ and $\Delta_{\mathcal{R}}(u_S, u_R) > \frac{1}{13}$.

Fix any $\varepsilon > 0$. We perturb players' payoffs in Example 4 to construct the desired positive share of environments. Similar to Section B.3.1, the idea is to show that changes to the persuasion payoff,

u_S	a_1	a_2	a_3
ω_1	$1/3 + s_{11}$	$2/3 + s_{12}$	$0 + s_{13}$
ω_2	$5/6 + s_{21}$	$1/6 + s_{22}$	$1/2 + s_{23}$

u_R	a_1	a_2	a_3
ω_1	$3/4 + r_{11}$	$1/2 + r_{12}$	$0 + r_{13}$
ω_2	$0 + r_{21}$	$1/2 + r_{22}$	$3/4 + r_{23}$

Table 5: Sender's and Receiver's payoffs

partitional persuasion payoff, and cheap-talk payoff are small under small perturbations. Since in Example 4, the value of commitment is zero and the value of randomization is positive, we obtain a positive measure of environments with a small value of commitment and a value of randomization bounded away from zero.

Players' payoffs are as in Table 5, where s_{ij}, r_{ij} are the perturbations to Sender's and Receiver's payoffs, respectively, when action a_j is taken in state ω_i.

Let $s_{ij}, r_{ij} \in [0, \delta]$, where $\delta > 0$. These perturbations generate a positive measure set of environments, denoted by E^δ. Consider any $(u_S, u_R) \in E^\delta$ with $\delta < 1/8$. Receiver's best response is

$$a_R(\mu) = \begin{cases} a_1 & \text{if } \mu \in [0, \mu_{12}] \\ a_2 & \text{if } \mu \in [\mu_{12}, \mu_{23}] \\ a_3 & \text{if } \mu \in [\mu_{23}, 1] \end{cases}$$

where $\mu_{12} = \frac{1/4 + r_{11} - r_{12}}{3/4 + r_{11} + r_{22} - r_{12} - r_{21}}$ and $\mu_{23} = \frac{1/2 + r_{12} - r_{13}}{3/4 + r_{23} - r_{13} + r_{12} - r_{22}}$. Note that $0 < \mu_{12} < 1/2 < \mu_{23} < 1$. Similar to Example 4, the optimal information structure induces beliefs μ_{12} and 1, yielding a value

$$\frac{1}{2(1-\mu_{12})} \max\{\mu_{12}(5/6 + s_{21}) + (1-\mu_{12})(1/3 + s_{11}), \mu_{12}(1/6 + s_{22}) + (1-\mu_{12})(2/3 + s_{12})\} + \frac{1 - 2\mu_{12}}{2(1-\mu_{12})}(1/2 + s_{23}).$$

Since $s_{ij} \in [0, \delta]$, taking $\delta \to 0$, we have $\mu_{12} \to 1/3$, and the above value approaches $1/2$. By continuity, there exists $\delta^1 > 0$ such that for any $\delta < \delta^1$, the persuasion value lies within the interval $[\frac{1}{2} - \frac{\varepsilon}{2}, \frac{1}{2} + \frac{\varepsilon}{2}]$.

Meanwhile, full revelation yields a payoff of $\frac{5/6+s_{11}+s_{23}}{2}$ and providing no information yields a payoff of $\frac{5/6+s_{12}+s_{22}}{2}$. Both values approach 5/12 when $\delta \to 0$. By continuity, there exists $\delta^2 > 0$ such that for any $\delta < \delta^2$, the partitional persuasion value is less than $\frac{5}{12} + \frac{\varepsilon}{2}$.

Next, we will construct a cheap-talk equilibrium that yields a payoff close to 1/2 for small δ. Consider a strategy profile (σ, ρ) with two on-path messages m_1, m_2: $\sigma(m_1|\omega_2) = \frac{\mu_{12}}{1-\mu_{12}}$, $\sigma(m_2|\omega_2) = \frac{1-2\mu_{12}}{1-\mu_{12}}$, $\sigma(m_1|\omega_1) = 1$; $\rho(a_1|m_1) = p$, $\rho(a_2|m_1) = 1-p$, $\rho(a_3|m_2) = 1$, where $p = \frac{1/3+s_{23}-s_{22}}{2/3+s_{21}-s_{22}}$. For any $\delta < 1/3$, $p \in (0,1)$ is a well-defined probability.

The strategy profile satisfies R-BR because the posterior upon observing m_1 is μ_{12}, and upon observing m_2 is 1. We now show that the profile also satisfies S-BR. For type ω_2 Sender, the expected payoff of sending message m_2 is $1/2 + s_{23}$ and the expected payoff of sending message m_1 is $p(5/6 + s_{21}) + (1-p)(1/6 + s_{22}) = 1/2 + s_{23}$, so type ω_2 Sender is indifferent and has no incentive to deviate. For type ω_1 Sender, the expected payoff of sending message m_2 is $0 + s_{13}$ and the expected payoff of sending message m_1 is $p(1/3 + s_{11}) + (1-p)(2/3 + s_{12})$. As $\delta \to 0$, $p(1/3+s_{11}) + (1-p)(2/3+s_{12}) \to 1/2$ and $0 + s_{13} \to 0$, so type ω_1 Sender strictly prefers to send message m_1. By continuity, there exists $\delta^3 > 0$ such that for any $\delta < \delta^3$, the strategy profile (σ, ρ) is a cheap-talk equilibrium, yielding a value $\frac{1}{2}(1/2+s_{23}) + \frac{1}{2}[p(1/3+s_{11})+(1-p)(2/3+s_{12})] > 1/2 - \frac{\varepsilon}{2}$.

Therefore, for any $\delta < \delta^* := \min\{\delta^1, \delta^2, \delta^3, \frac{1}{8}\}$, and for any $(u_S, u_R) \in E^\delta$, $\Delta_{\mathcal{C}}(u_S, u_R) < (\frac{1}{2} + \frac{\varepsilon}{2}) - (\frac{1}{2} - \frac{\varepsilon}{2}) = \varepsilon$, and $\Delta_{\mathcal{R}}(u_S, u_R) > (\frac{1}{2} - \frac{\varepsilon}{2}) - (\frac{5}{12} + \frac{\varepsilon}{2}) = \frac{1}{12} - \varepsilon$.

We have established that for any $\varepsilon > 0$, there exists a positive measure set of environments E such that for any $(u_S, u_R) \in E$, we have $\Delta_{\mathcal{C}}(u_S, u_R) < \varepsilon$ and $\Delta_{\mathcal{R}}(u_S, u_R) > \frac{1}{12} - \varepsilon$. Now fix any $\hat{\varepsilon} > 0$ and let $\varepsilon = \min\{\hat{\varepsilon}, \frac{1}{12} - \frac{1}{13}\}$. Our previous result implies that there exists a positive measure set of environments E such that for any $(u_S, u_R) \in E$, $\Delta_{\mathcal{C}}(u_S, u_R) < \varepsilon \leq \hat{\varepsilon}$ and $\Delta_{\mathcal{R}}(u_S, u_R) > \frac{1}{12} - \varepsilon \geq \frac{1}{12} - (\frac{1}{12} - \frac{1}{13}) = \frac{1}{13}$.

B.4 Open set of environments for the proof of Proposition 1

B.4.1 An open set of environments containing Example 1

We construct an open set of environments in which $V^{BP} = V^{MD} > V^{CT} = V^{babbling}$.

Consider $\Omega = \{0, 1\}$ with equal prior, and $A = \{a_1, a_2, a_3\}$.

Let $r_{ij} := u_R(a_j, \omega = i)$ and $s_{ij} := u_S(a_j, \omega = i)$ for $i \in \{0, 1\}$ and $j \in \{1, 2, 3\}$. Consider any

environment satisfying the following:

$$r_{01} > r_{02} > r_{03}, \qquad r_{13} > r_{12} > r_{11},$$

$$r_{12} - r_{11} > r_{01} - r_{02}, \qquad r_{02} - r_{03} > r_{13} - r_{12},$$

$$s_{02} > s_{03} > s_{01}, \qquad s_{13} > s_{12} > s_{11},$$

$$(r_{12} - r_{11})(s_{02} - s_{03}) > (r_{01} - r_{02})(s_{13} - s_{12}).$$

Since the constraints are all strict inequalities, they define an open set of environments. In addition, the set is non-empty because the payoffs in Example 1 satisfy these constraints.

Define $\mu_{12} := \frac{r_{01}-r_{02}}{(r_{01}-r_{02})+(r_{12}-r_{11})}$ and $\mu_{23} := \frac{r_{02}-r_{03}}{(r_{02}-r_{03})+(r_{13}-r_{12})}$ as Receiver's cutoff beliefs between actions a_1, a_2 and a_2, a_3. By the assumptions, $0 < \mu_{12} < \frac{1}{2} < \mu_{23} < 1$. Hence, Receiver's best response is a_1 for $\mu \leq \mu_{12}$, a_2 for $\mu \in [\mu_{12}, \mu_{23}]$, and a_3 for $\mu \geq \mu_{23}$.

Mediation Consider the following action recommendation:

$$\pi(a_2|\omega = 0) = 1, \quad \pi(a_2|\omega = 1) = \frac{\mu_{12}}{1 - \mu_{12}}, \quad \pi(a_3|\omega = 1) = \frac{1 - 2\mu_{12}}{1 - \mu_{12}}.$$

This recommendation scheme is obedient because the posterior when seeing recommendation a_2 is μ_{12}, and the posterior when seeing recommendation a_3 is 1.

It also satisfies Sender's truth-telling constraint. Note that Sender prefers action a_2 in state $\omega = 0$ and prefers action a_3 when $\omega = 1$. In state $\omega = 1$, truthful reporting leads to a mixture of a_2 and a_3, while misreporting leads to a_2. In state $\omega = 0$, truthful reporting leads to action a_2, while misreporting leads to a mixture of a_2 and a_3.

The recommendation scheme yields a Sender's payoff of $\frac{1}{2}\big(s_{02} + \frac{\mu_{12}}{1-\mu_{12}} s_{12} + \frac{1-2\mu_{12}}{1-\mu_{12}} s_{13}\big)$. We will next establish that this is the persuasion value.

Persuasion Let $v_i(\mu) = \mu s_{1i} + (1-\mu)s_{0i}$ for $i = 1, 2, 3$. Sender's indirect utility function (with ties broken in favor of Sender) can then be written as

$$V(\mu) = \begin{cases} v_1(\mu) & \text{if } \mu < \mu_{12} \\ v_2(\mu) & \text{if } \mu \in [\mu_{12}, \mu_{23}) \\ v_3(\mu) & \text{if } \mu > \mu_{23} \\ \max\{v_2(\mu_{23}), v_3(\mu_{23})\} & \text{if } \mu = \mu_{23}. \end{cases}$$

Let ℓ be the affine function through $(\mu_{12}, v_2(\mu_{12}))$ and $(1, v_3(1))$. Since $(r_{12} - r_{11})(s_{02} - s_{03}) > (r_{01} - r_{02})(s_{13} - s_{12})$, we have $v_2(\mu_{12}) > v_3(\mu_{12})$. Since v_3 is an affine function, $\ell(1) = v_3(1)$ and $\ell(\mu_{12}) = v_2(\mu_{12}) > v_3(\mu_{12})$, we have $\ell(\mu) \geq v_3(\mu)$ for all $\mu \in [0, 1]$. Since $s_{03} > s_{01}$ and $s_{13} > s_{11}$, $v_3 \geq v_1$ and so $\ell(\mu) \geq v_1(\mu)$ for all $\mu \in [0, 1]$.

Since $s_{13} > s_{12}$, $v_3(1) > v_2(1)$. Since v_2 is an affine function, $\ell(\mu_{12}) = v_2(\mu_{12})$ and $\ell(1) = v_3(1) > v_2(1)$, we have $\ell(\mu) \geq v_2(\mu)$ for all $\mu \in [\mu_{12}, 1]$.

Therefore, we have established that $V(\mu) \leq \ell(\mu)$ for all μ. This implies that

$$\begin{aligned} cav(V)(\mu_0) &\leq \ell(\mu_0) \\ &= \frac{1}{2(1-\mu_{12})} v_2(\mu_{12}) + \frac{1-2\mu_{12}}{2(1-\mu_{12})} s_{13} \\ &= \frac{1}{2(1-\mu_{12})} (\mu_{12} s_{12} + (1-\mu_{12}) s_{02}) + \frac{1-2\mu_{12}}{2(1-\mu_{12})} s_{13} \\ &= \frac{1}{2}\big(s_{02} + \frac{\mu_{12}}{1-\mu_{12}} s_{12} + \frac{1-2\mu_{12}}{1-\mu_{12}} s_{13}\big). \end{aligned}$$

Therefore,

$$V^{BP} = V^{MD} = \frac{1}{2}\big(s_{02} + \frac{\mu_{12}}{1-\mu_{12}} s_{12} + \frac{1-2\mu_{12}}{1-\mu_{12}} s_{13}\big).$$

Cheap Talk Note that the babbling equilibrium yields a Sender's payoff of $\frac{s_{02}+s_{12}}{2}$. We will show that no cheap-talk equilibrium can yield a strictly higher payoff.

Suppose toward a contradiction that there is a cheap-talk equilibrium with Sender payoff strictly above the babbling payoff $\frac{s_{02}+s_{12}}{2}$. Let U_ω denote Sender's equilibrium payoff in state ω. Since $U_0 \leq s_{02}$, we must have $U_1 > s_{12}$. This means that every message sent in state 1 must give a payoff

strictly above s_{12}. Since only action a_3 yields a payoff strictly above s_{12} in state 1, it must be induced with strictly positive probability under all such messages. Therefore, the induced posterior must be greater than $\mu_{23} > 1/2$.

By Bayes plausibility, since the prior is $1/2$, there must be an on-path message inducing posterior below $1/2$. Such a message cannot be sent in state 1, because it gives a payoff no more than s_{12}. So it is sent only in state 0, which induces a posterior 0 and leads to action a_1. Hence, $U_0 = s_{01}$.

Now take any message sent in state 1, which must induce a_2 with probability p and a_3 with probability $1-p$ for some $p \in [0,1)$. If state 0 Sender sends this message, his expected payoff will be $ps_{02} + (1-p)s_{03} > s_{01}$. This contradicts $U_0 = s_{01}$. Hence, no cheap-talk equilibrium yields more than the babbling payoff.

Therefore, for the constructed open set of environments,

$$V^{BP} = V^{MD} > V^{CT} = V^{babbling}.$$

B.4.2 An open set of environments containing Example 2

We construct an open set of environments in which $V^{MD} = V^{CT} > V^{\text{partitional } CT}$.

Consider $\Omega = \{0,1\}$ with equal prior, and $A = \{a_1, a_2, a_3\}$.

Let $r_{ij} := u_R(a_j, \omega = i)$ and $s_{ij} := u_S(a_j, \omega = i)$ for $i \in \{0,1\}$ and $j \in \{1,2,3\}$. Consider any environment satisfying the following:

$$\begin{aligned}
&r_{01} > r_{02} > r_{03}, \qquad r_{13} > r_{12} > r_{11},\\
&r_{12} - r_{11} > r_{01} - r_{02}, \qquad r_{02} - r_{03} > r_{13} - r_{12},\\
&s_{03} > s_{01} > s_{02}, \qquad s_{13} > s_{12} > s_{11}.
\end{aligned}$$

Since the constraints are all strict inequalities, they define an open set of environments. In addition, the set is non-empty because the payoffs in Example 2 satisfy these constraints.

Define

$$\mu_{12} := \frac{r_{01} - r_{02}}{(r_{01} - r_{02}) + (r_{12} - r_{11})}$$

and

$$\mu_{23} := \frac{r_{02} - r_{03}}{(r_{02} - r_{03}) + (r_{13} - r_{12})}$$

as Receiver's cutoff beliefs between actions a_1, a_2 and a_2, a_3. By the assumptions,

$$0 < \mu_{12} < \frac{1}{2} < \mu_{23} < 1.$$

Hence Receiver's best response is a_1 for $\mu \leq \mu_{12}$, a_2 for $\mu \in [\mu_{12}, \mu_{23}]$, and a_3 for $\mu \geq \mu_{23}$.

Define

$$p := \frac{s_{03} - s_{01}}{s_{03} - s_{02}}.$$

Since $s_{03} > s_{01} > s_{02}$, we have $p \in (0, 1)$, and

$$ps_{02} + (1 - p)s_{03} = s_{01}.$$

Partitional Cheap Talk With a binary state, a partitional cheap-talk profile is either full revelation or babbling.

Full revelation is not an equilibrium. Under full revelation, the message sent in state $\omega = 0$ induces a_1 and the message sent in state $\omega = 1$ induces a_3. But type $\omega = 0$ strictly prefers deviating to the $\omega = 1$ message, because $s_{03} > s_{01}$.

Babbling induces action a_2, yielding a Sender's payoff $\frac{1}{2}s_{02} + \frac{1}{2}s_{12}$. Therefore,

$$V^{\text{partitional } CT} = V^{babbling} = \frac{1}{2}s_{02} + \frac{1}{2}s_{12}.$$

Cheap Talk We now construct a cheap-talk equilibrium. Consider the following strategy profile with two on-path messages m_1 and m_2: Sender's strategy is $\sigma(m_2 \mid \omega = 1) = 1$, $\sigma(m_2 \mid \omega = 0) = \frac{1-\mu_{23}}{\mu_{23}}$, and $\sigma(m_1 \mid \omega = 0) = \frac{2\mu_{23}-1}{\mu_{23}}$. Receiver's strategy is $\rho(a_1|m_1) = 1$, $\rho(a_2|m_2) = p$, and $\rho(a_3|m_2) = 1 - p$.

The profile satisfies R-BR because when seeing m_1 Receiver's posterior is 0 and when seeing m_2 Receiver's posterior is μ_{23}. We verify that the profile also satisfies S-BR. Type $\omega = 0$ Sender gets s_{01} from sending m_1, and gets $ps_{02} + (1-p)s_{03} = s_{01}$ from sending m_2. Hence type $\omega = 0$ is indifferent

and is willing to mix. Type $\omega = 1$ Sender gets s_{11} from sending m_1, and gets $ps_{12} + (1-p)s_{13}$ from sending m_2. Since $s_{13} > s_{12} > s_{11}$, type $\omega = 1$ strictly prefers to send m_2.

Therefore, the strategy profile is a cheap-talk equilibrium, yielding a Sender's payoff of

$$\frac{1}{2}s_{01} + \frac{1}{2}\big[ps_{12} + (1-p)s_{13}\big],$$

which is strictly greater than the babbling payoff because $s_{01} > s_{02}$ and $s_{13} > s_{12}$.

Thus

$$V^{CT} > V^{\text{partitional } CT}.$$

Mediation It remains to show that mediation cannot improve on the payoff achieved by the cheap-talk equilibrium constructed above.

Suppose $\pi : \Omega \to \Delta A$ is induced by some mediated equilibrium, which must satisfy obedience and truth-telling. The obedience constraints for actions a_2 and a_3 give

$$(1-\mu_{23})\pi(a_2|\omega = 1) \leq \mu_{23}\pi(a_2|\omega = 0)$$

and

$$(1-\mu_{23})\pi(a_3|\omega = 1) \geq \mu_{23}\pi(a_3|\omega = 0).$$

Now consider Sender's truth-telling constraint in state $\omega = 0$. Truthful reporting gives type $\omega = 0$ payoff

$$s_{01} + (s_{02} - s_{01})\pi(a_2|\omega = 0) + (s_{03} - s_{01})\pi(a_3|\omega = 0)$$

while misreporting as type $\omega = 1$ gives

$$s_{01} + (s_{02} - s_{01})\pi(a_2|\omega = 1) + (s_{03} - s_{01})\pi(a_3|\omega = 1).$$

Truth-telling requires

$$(s_{02} - s_{01})\pi(a_2|\omega = 0) + (s_{03} - s_{01})\pi(a_3|\omega = 0) \geq (s_{02} - s_{01})\pi(a_2|\omega = 1) + (s_{03} - s_{01})\pi(a_3|\omega = 1).$$

Since $s_{02} - s_{01} < 0$ and $s_{03} - s_{01} > 0$, Receiver obedience implies

$$(s_{02}-s_{01})\pi(a_2|\omega=0)+(s_{03}-s_{01})\pi(a_3|\omega=0) \leq \frac{1-\mu_{23}}{\mu_{23}}\Big[(s_{02}-s_{01})\pi(a_2|\omega=1)+(s_{03}-s_{01})\pi(a_3|\omega=1)\Big].$$

Since $\frac{1-\mu_{23}}{\mu_{23}} < 1$, combining the previous two inequalities yields

$$(s_{02} - s_{01})\pi(a_2|\omega=1) + (s_{03} - s_{01})\pi(a_3|\omega=1) \leq 0.$$

Therefore, type $\omega = 0$'s payoff under π is at most s_{01}.

We now bound type $\omega = 1$'s payoff. The previous inequality is equivalent to

$$\pi(a_3|\omega=1) \leq \frac{s_{01} - s_{02}}{s_{03} - s_{01}}\pi(a_2|\omega=1).$$

Together with $\pi(a_2|\omega=1) + \pi(a_3|\omega=1) \leq 1$, this gives

$$\pi(a_3|\omega=1) \leq \frac{s_{01} - s_{02}}{s_{03} - s_{02}} = 1 - p.$$

Therefore type $\omega = 1$'s payoff is bounded above by

$$\begin{aligned}
&s_{11}\pi(a_1|\omega=1) + s_{12}\pi(a_2|\omega=1) + s_{13}\pi(a_3|\omega=1)\\
\leq& s_{12}[\pi(a_1|\omega=1) + \pi(a_2|\omega=1)] + s_{13}\pi(a_3|\omega=1)\\
=& s_{12}\big(1 - \pi(a_3|\omega=1)\big) + s_{13}\pi(a_3|\omega=1)\\
=& s_{12} + (s_{13} - s_{12})\pi(a_3|\omega=1)\\
\leq& s_{12} + (s_{13} - s_{12})(1-p)\\
=& s_{12}p + s_{13}(1-p).
\end{aligned}$$

The first inequality follows from $s_{12} > s_{11}$, and the second inequality follows from $s_{13} > s_{12}$ and $\pi(a_3|\omega=1) \leq 1 - p$.

Therefore, any mediated equilibrium gives Sender at most a payoff of

$$\frac{1}{2}s_{01} + \frac{1}{2}[s_{12}p + s_{13}(1-p)],$$

which is the payoff achieved in the previous cheap-talk equilibrium. Therefore, for the constructed open set of environments,

$$V^{MD} = V^{CT} > V^{\text{partitional } CT}.$$